

”Montreal Forced Aligner applied to a Low-Resource Consonant-heavy Language”

Abstract

In this research we have explored the challenges when doing Forced Alignment with an MFA while working with low-resource languages, and the issues that one may encounter. The target language in particular was a Berber language: Tachelhit (or Shilha, iso-code: **shi**). Berber languages have different levels of online representation, and **shi** in particular has few reliable sources for audio. In this thesis I document several datasets and the issues found while trying to use them. While the writing for Berber languages is relatively standardized, the pronunciation of different dialects is not so well-established. During this thesis I cross-compare the various resources that exist to create an orthography that allows transforming text to IPA in **shi/tzm** and separately **kab** to IPA. To improve the MFA's performance, **kab**, another Berber language with far more ASR resources, is used as the base model to later adjust (finetune) to **zgh** (a standard form of **shi**). My analysis shows using **kab** as a footstone for better Forced Alignment on **shi** to be advisable, and it being specially critical when aligning on a different dialect, **tzm**, almost-unseen by the model, which is part of the **zgh** standard (encompassing both **tzm/shi**). Some further analysis is done comparing phone-level and word-level alignment.

1 Introduction

MENTION LACK OF RESEARCH LACK OF KNOWLEDGE (CONSENSUS/UNDERSTANDING) LIMITATIONS OF PREVIOUS STUDIES PRACTICAL PROBLEM

WHY THIS IS THE FIRST STUDY TO DO X, TIE WITH KNOWLEDGE GAP

A Montreal Forced Aligner (MFA moving forward) [1] is a model that time-aligns text annotations for an audio file. MFA as a technology has a rich amount of documentation ¹ as well as support for a wide variety of languages ². The pipeline for this technology is quite flexible depending in the resources available ³; but can be roughly summarized as combining a series of transcripts, a pronunciation dictionary and an acoustic model to perform word timestamp annotation on a series of transcripts. Figure 1 shows us how some of how to get each of these individual components. It must be noted that MFA is completely symbol-agnostic: the symbols in a word can be arbitrary, as well as the symbols for its pronunciation; each symbol for pronunciation may have multiple realizations in itself (especially depending on context).

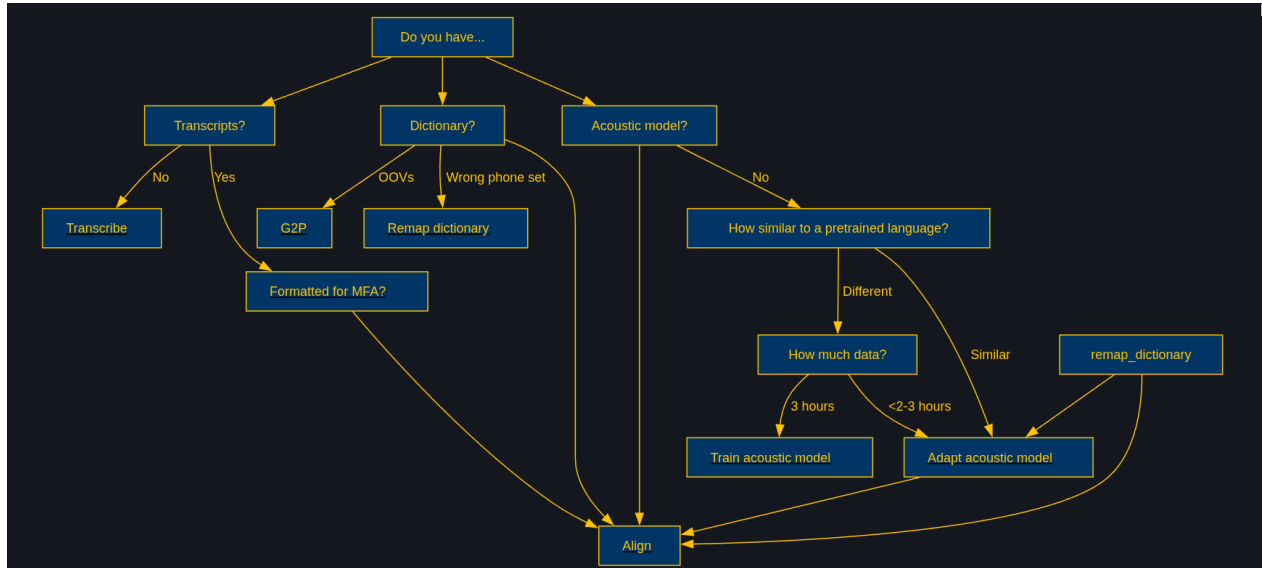


Figure 1: Requirement flowchart from https://montreal-forced-aligner.readthedocs.io/en/latest/first_steps/index.html#first-steps-adapt-pretrained

MFA builds on top of Kaldi [2]⁴ which uses a combination of Gaussian Mixture models (for each state, a GMM processes the current phone) and Hidden Markov Models (HMM govern the transition between one phone and the next). Generally speaking, every window of frames, the model questions whether they belong to the current phone or one of the possible following phones (a symbol may have multiple realizations, Figure 2 gives a rough idea). What MFA adds on top of Kaldi are mostly "convenience" features: pre-trained acoustic models, much simpler data preparation, "Out of vocabulary" detection and handling, friendly ways to expand pronunciation dictionaries, etc.

¹<https://montreal-forced-aligner.readthedocs.io/>

²<https://mfa-models.readthedocs.io/en/latest/>

³https://montreal-forced-aligner.readthedocs.io/en/latest/first_steps/index.html#first-steps

⁴<https://kaldi-asr.org>

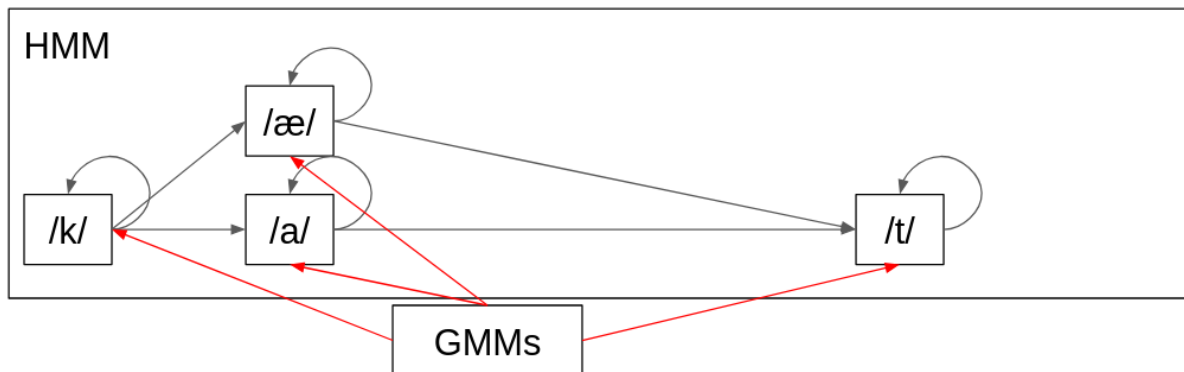


Figure 2: Example of "cat" pronunciation in the Kaldi pipeline. Given a window: The HMM may transition from a state to the same (if the current phone is more likely than any of the following) or to following state, depending on which probability is higher. The probability of each phone is calculated using GMMs. Based on APX Machine Learning's course for Speech recognition: <https://apxml.com/courses/introduction-to-speech-recognition/chapter-3-acoustic-modeling/combining-gmms-and-hmms>

The pipeline of training each aligner model ⁵ roughly functions in this manner:

- **1. Monophone Modelling** Each individual phone is modeled (using Gaussian Mixture Models)
- **2. Triphone Modelling** Each individual phone is modeled but the preceding and upcoming phones are added. Phone grouping is possible here (e.g. /n/ and /ŋ/ can be clustered).
- **3. LDA + MLLT** Dimensionality remapping of phone data in order to maximize the differences between different phones (similar to PCA, but instead of maximizing variance these techniques are used to maximize separation between phones). Phone groups are relevant.
- **4. Speaker differentiation** Where each speaker gets their MFCC calculated. Unexplored in this project (single-speaker).

For a more realistic approach to dialectal variety, one may specify a series of replacement rules ⁶(to take into account alternative realizations, spirantizations or assimilations) and phone grouping ⁷(for phonemes that are similar). Replacement rules are applied prior to Monophone Modelling, whereas Phone Groups are used in every step past Monophone Modelling. Replacement rules only may be applied intra-word (so cross-word interactions between phones must be modelled as /phone+end_of_word/ may become /different_phone/, where different_phone may have multiple realizations).

⁵Explained at https://montreal-forced-aligner.readthedocs.io/en/latest/user_guide/index.html#user-guide

⁶https://montreal-forced-aligner.readthedocs.io/en/latest/user_guide/implementations/phonological_rules.html

⁷https://montreal-forced-aligner.readthedocs.io/en/latest/user_guide/implementations/phone_groups.html

Berber languages (also known as Tamazight or Amazigh languages) are a series of mostly-spoken languages spoken by native communities in North Africa, belonging to the Afro-Asiatic tree (Figure 3). These languages, that historically belonged only to the indigenous communities, have gained officiality recently in some countries (Morocco in 2011 [3], Algeria in 2016 [4]). This has lead to them becoming standardized to some degree [5] as well as receiving UNICODE support for their writing [6].

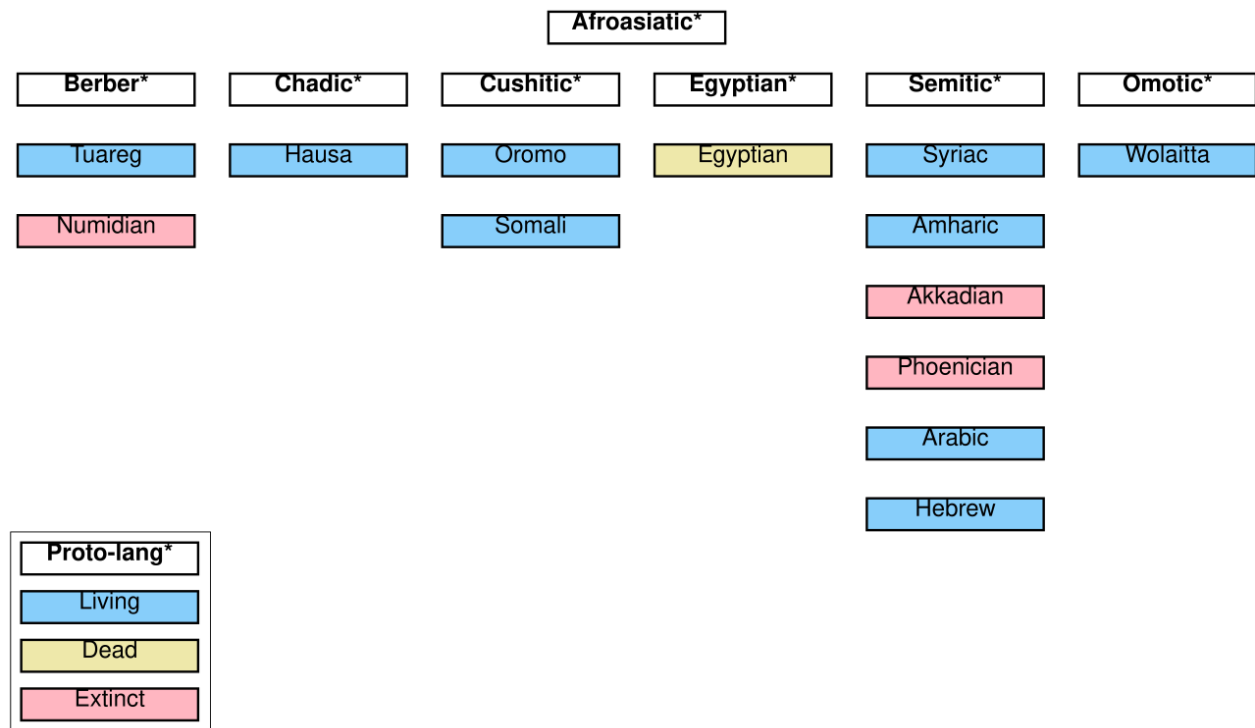


Figure 3: Afroasiatic family tree from https://en.wikipedia.org/wiki/Afroasiatic_languages

Culturally, Berber languages have gotten some recent attention (in contrast to the past, when Arabic was exclusively the vehicular language in the countries where Berber Languages are present) [7] and grammar books for them have been worked on since a long time [8] in an effort to raise their cultural relevance.

In Figure 4 the Ontology for the languages involved in this Thesis is roughly described. Most of the Berber languages makes a continuum rather a proper family tree (**tzm** is a dialect continuum between **shi** and **rif**). Intelligibility can be quite arbitrary, but Berber speakers of adjacent areas tend to understand each other.

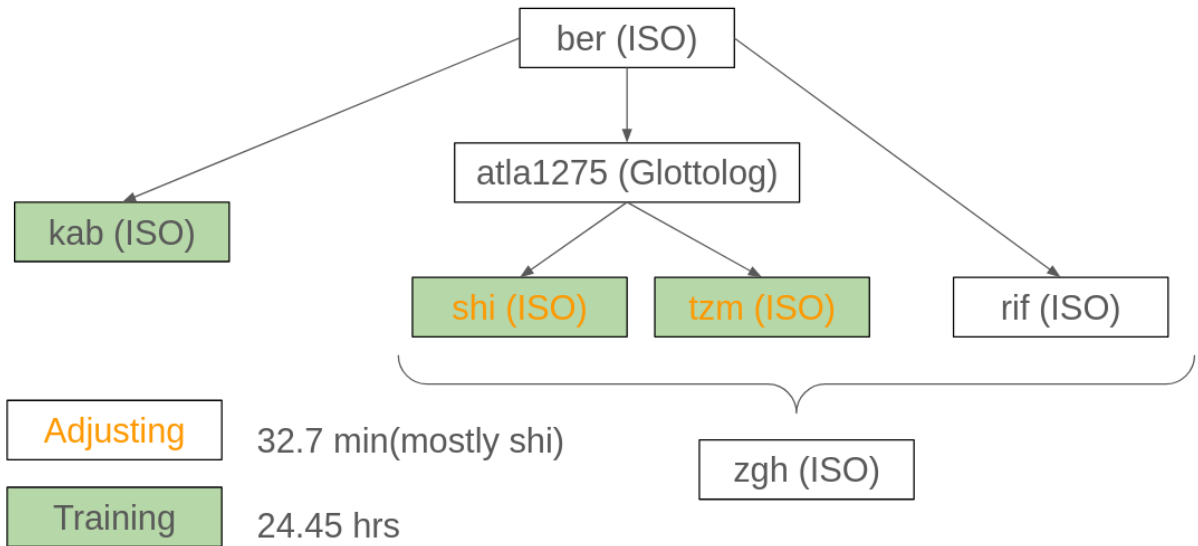


Figure 4: Ontology describing the Berber languages relevant for this Thesis based on Wikipedia. **ber** stands for Berber/Tamazight/Amazigh Languages, **kab** for kabyle, **shi** for Shilha/Tachelhit, **tzm** for Central Atlas Tamazight, **rif** for Rifian/Tarifit, **zgh** for Standard Moroccan Amazigh. A rough estimate for the corpus available from common_voice_22.0 is written on the left. **zgh** is a standard form combining **shi**, **tzm** and **rif**.

Each Berber language is typically written in one of the 3 alphabets (Figures 5, 6 and 7) and sometimes multiples of them. Different alphabets are normally not used heterogenously within a text. Some alphabets may include or exclude symbols to accomodate the specific variant. For our relevant languages: **kab** uses Latinscript (with marginal Tifinagh usage), **tzm** uses Arabscript and Tifinagh, **zgh** uses Tifinagh and **shi** uses all 3 alphabets.

A huge amount of **kab** corpus is readily available online; but the same cannot be said about the rest of Berber languages. Please note that Tamazight/Amazigh is often used for the **tzm** as well as the overarching **ber**; making it necessary to double-check when one finds resources labeled as Tamazight/Amazigh. Unfortunately I have not been able to find reliable corpus to evaluate or align in **rif** (one of the Moroccan Berber variants). It is included in the Ontology nonetheless, as **zgh** is a Standardized version of the Moroccan Amazigh majoritary varieties (**shi**, **tzm**, **rif**).

A B C D E F G H I
J K L M N O P Q
R S T U V W X Y Z

*a b c d e f g h i j k l m n
o p q r s t u v w x y z*

Figure 5: Latin script.

ا ب ت ث ج ح خ
kh ḥ j th t ḅ '

د ذ ر ز س ش ص
ṣ sh s z r dh d

ض ط ظ ع غ ف ق
q f gh ' z t ḍ

ك ل م ن ه و ي
y w h n m l k

Figure 6: Arabic script.

ⵜ ⵍ ⵎ ⵏ ⵙ ⵔ ⵖ ⵓ ⵙ

Figure 7: Tifinagh.

Since the first negotiations of the Thesis topic, Tachelhit (also known as Shilha, with isocode **shi**) was identified as a good target for a low-resource language. Some amount of written/spoken corpus can be found, but definitely limited. A lot of theoretical literature exists [9]; but some cross-referencing with public datasets is due. **shi** also has a series of very exotic properties, that will be explained in the following sections.

MFA allows for training in a language and then adjust the trained model to a different language. During this thesis, the implementation objective was to use **kab** for base training (since there is more corpus than I could realistically use) and adjust the **kab** model (one could say fine-tune) using the **zgh** corpus (and test using the longest samples of **shi/tzm** respectively).

2 Motivation

The focus of this Thesis is particularly on **shi**. As mentioned, this language happens to be low resource, have relatively similar languages with enormous corpus and plenty of literature describing it. Compared to Indo-European languages it is definitely unusual. The literature [9] has examples, which help me further illustrate further how Tachelhit is an outlier among languages:

Latinized **shi**: **kksttšštt**

IPA: /k:st:š:t:/

Translation: *remove it (m) and eat it (m)*

Latinized **shi**: **tfttštskkxtkkstšškk**

IPA: /tft:šts:k:χtk:stš:k:/

Translation: *you examined the currency which you checked out*

As is evident by both sentences, Tachelhit is a consonant-heavy language. While vowels exist in the language and are relatively common; sentences with no vowels are completely acceptable. Readers more familiar with how "š" represents the IPA /ʃ/, may have noticed another particularity about these sentences: they are entirely voiceless, they should be pronounced with no vocal chord vibration; voiceless sentences are also acceptable in **shi**.

Another set of particularities from this language are regarding its limited vowels: vocalic sounds often appear inbetween the consonants in the language without lexically contrasting words or explicit writing. Besides these, an epenthetic schwa "e/ə" is written (that may be pronounced or not) and the limited set of vowels ("a","u","i") have relatively wide realizations with no lexical contrastive difference (likely as a result of the consonantic focus of the language).

The combination of these factors make **shi** a clear outlier in comparison to the more "vocalic" languages that most Forced Aligner algorithms are designed to cover.

Lastly, while the language writing is decently standardized, some quirks exist with pronunciation. An objective in this thesis is to provide a good pronunciation standard working across **zgh** variants for orthographical transliteration. One would guess that given how Berber languages pronunciation tends to follow a one-character-one-sound pattern, this would result in a simple universal phonological space. When looking at the phonological space for each variant; this is clearly not the case. Assimilations (spirantizations, devoicing, voicing) are not so standardized and non-standard symbols exist. As will be clear by the Section 4, the phonological space and alphabet do not match as easily as one would wish.

3 Related work

Research suggests that usage of preexisting model languages results in more precise boundary detection for phones; as well as the usage of multiple languages when training [10]. These suggestions match the ones done by the MFA maintainer [11], let me quote him in regards to how many audio hours are needed when training a model:

"[...] The more nuanced answer is that when there's a pretrained model on 20+ hours of speech similar to the variety you're trying to align, that will likely generate pretty good alignments[...]"

In contrast, the documentation suggests 1000 hours⁸. Nonetheless, for the scope of this thesis we will hope that the roughly 25 **kab** hours suffice.

Although at the end **common_voice_22.0** has been exclusively used for this Thesis, I will list all the datasets for speech recognition in Berber languages that were considered at some point. They reflect the challenges faced when working with a low-resource language:

- **Tamazight-Arabic Speech Recognition Dataset [12]:** A corpus of 15.5 hours generated through youtube videos (specially Tamazight movies) particularly focused on the Tachelhit dialect. It is also worth noting the method for generating this dataset is programmatically reproducible⁹ and that the original video subtitles are manually annotated in Arabscript (which are then then transcribed with video recognition using screenshots¹⁰). Unfortunately, this dataset's text is in Arabic language, not Amazigh in Arabic script; therefore unusable for our Forced Aligner. This was easy to misunderstand; as it mentions "speech segments in Tamazigh [...] with corresponding Arabic transcriptions". Nonetheless the error of initially considering it as valid is on me.
- **Tamazight dataset from Tifinlab [13]:** an estimate of 10 hours (supposedly 9 of Kabyl¹¹ and 1 of Moroccan Amazigh¹²) of latin-based Berber orthography (Latin Script). Creative Commons licensed. Upon consulting with 2 native speakers^{13,14}; they both confirmed that the Moroccan Amazigh samples (the most important) was mislabeled and it contained Kabyle. Given its unreliability, I decided to drop both the Moroccan Amazigh and Kabyl dataset (plenty of validated **kab** data exists already).
- **AMZSRD [14]** This dataset contains 520 words pronounced by 50 speakers of ages 18-40 repeating each word 5 times, for a grand total of 130000 files. I contacted the maintainer of this dataset during this Thesis' realization, receiving no response. While these would have been helpful as training examples (for phones in **rif**), and would enable interesting experimentation; they would not have been robust evaluation for Forced Alignment, given how at best we could combine single-word utterances into a sentence.

⁸https://montreal-forced-aligner.readthedocs.io/en/latest/user_guide/troubleshooting.html#errors-training-on-small-corpora

⁹<https://github.com/Dahimi/Tamazight-ASR>

¹⁰https://github.com/Dahimi/Tamazight-ASR/blob/main/tamazight_asr/prototyping/transcripts_extraction.ipynb

¹¹https://huggingface.co/datasets/TutlayAI/kabyle_asr

¹²https://huggingface.co/datasets/TutlayAI/moroccan_amazigh_asr

¹³<https://www.linkedin.com/in/abdelhaqueidali/>

¹⁴<https://www.linkedin.com/in/hichammellouki/>

- **Tamazight NLP TOSD Dataset [15]** 1.2 hours of a single Native Tachelhit speaker; one of the same speakers that helped me identify the issues with Tifinlab's Moroccan Amazigh dataset¹⁵. The speaker has a higher educational background in linguistics. Not included due to it being found rather late in the thesis development.
- **Common Voice 22_o [16]**: 1.31 validated hours of ¹⁶**zgh** in Tamazigh Script ¹⁷ as well as 570 **kab** validated hours ¹⁸. The distinction among **shi/tzm/rif** may be a point of debate in this dataset still¹⁹. Public Domain. The **zgh** dataset has limited variability (in both gender and age), as well as no distinction between casual vs clear speech style.

The data from **Common Voice 22_o [16]**: is community-validated, and having Tachelhit-specific audios that have gone through some curation makes them more valuable; this dataset was left as the exclusive training corpus and most of the evaluation. The initial intention was for **Common Voice 22_o [16]**'s **shi corpus** to be the testing (leaving the rest as training) due to it being validated, but Tifinlab's and Dahimi's dataset had to be discarded. Predictably the results for this dataset should yield higher accuracy compared to a real use case, as the guidelines ²⁰ are quite focused on ASR (clear pronunciation, no background noises).

¹⁵<https://www.linkedin.com/in/abdelhaqueidali/>

¹⁶https://huggingface.co/datasets/fsicoli/common_voice_22_o/blob/main/release_stats.py, "zgh"

¹⁷https://huggingface.co/datasets/fsicoli/common_voice_22_o/raw/main/transcript/zgh/train.tsv, the rows with **zgh/shi** are Tachelhit

¹⁸https://huggingface.co/datasets/fsicoli/common_voice_22_o/blob/main/release_stats.py, "kab"

¹⁹<https://discourse.mozilla.org/t/add-tamazigh-language/36659/6>

²⁰<https://commonvoice.mozilla.org/en/guidelines?tab=scripted-speech#public-domain>

A research article with an approach similar to my implementation exists for **kab** [17], where orthographic transliteration is used instead of a manually written pronunciation dictionary; however when validating the implementation I noticed that the dictionary used for transliteration ²¹ lacks depth in domain knowledge. Overall, the paper uses a previous version of Common Voice; but mixes up properties of **zgh** and **kab** in an arbitrary manner, this is specially bad when it comes to the particularities on **kab** geminates. A clear example is equating the Tifinagh symbol "ⵜ" to the latin script "tt". While this may be correct for **shi/tzm**, I can guarantee that does not hold for Kabyle ("tt" are read as /ts/). Hearing a native speaker and audios from Common Voice; one will notice also quickly that while the Latin script "b" can, depending on the preceding sound, be realized as a /b/ or /β/ ²² which are encoded as "Θ" and "⊕" respectively (the second symbol being rarely if ever used). Instead this source encodes "ⵇ" as the only way to encode "⊕". In my implementation ²³, I have chosen to keep **kab** and the different variants of **zgh** separate to avoid misrepresenting the IPA pronunciation. I decide this after a lot of cross-referencing. Kabyle also has a tendency on transforming geminates to stops: E.g.: /jj/ -> can be pronounced /j:/ or /g/; /kk/ can be pronounced /k:/ or /q/. The consonant overall results and the fact that vowelless words perform worse will be used in my result analysis, but should be taken with a grain of salt. These errors have not been noted in other research [18].

ASR reseach on **shi** [19] shows that the use of Arabic script (due to its more dissimilar phonologic inventory) results in noticeably lower performance for the more exotic properties of **shi** (voiceless words, specially in casual speaking style).

²¹available at https://github.com/berbertranslit/berbertranslit/blob/main/data/tifinagh_ahaggar/tifinagh_ahaggar.yml

²²https://en.wikipedia.org/wiki/Kabyle_language#Fricatives_vs._stops

²³available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/kab_build_dicts.py

4 Orthographical transliteration

In hopes of writing an aligner that works well for all **zgh** variants; we will stick to a relatively broad transcription; attempting to best capture the different consonants that are lexically contrastive and writing the realizations for the **shi/tzm** dialects.

	TIFINAGHE	Correspondance latine	Correspondance arabe	Exemples
ya	ⵢ	a	ا	ⵢⵏⵓⵢ
yab	ⵢⵓ	b	ب	ⵢⵓⵢⵓⵢ
yag	ⵢⵖ	g	ك	ⵢⵖⵓⵢ
yag [~]	ⵢⵖ [~]	g [~]	ك [~]	ⵢⵖⵖⵖⵖⵢ
yad	ⵢⵔ	d	د	ⵢⵔⵓⵢ
yaɖ	ⵢⵔ	ɖ	ض	ⵢⵔⵓⵢ
yey	ⵢⵉ	e		ⵢⵉⵢⵓⵢ
yaf	ⵢⵑ	f	ف	ⵢⵑⵓⵢ
yak	ⵢⵏ	k	ك	ⵢⵏⵓⵢⵓⵢ
yak [~]	ⵢⵏ [~]	k [~]	ك [~]	ⵢⵏⵏⵏⵏⵢⵓⵢ
yah	ⵢⵁ	h	ه	ⵢⵁⵢⵓⵢ
yah ^h	ⵢⵁ	h ^h	ح	ⵢⵁⵢⵓⵢ
yæ	ⵢⵏ	ɛ	ع	ⵢⵏⵓⵢ
yax	ⵢⵁ	x	خ	ⵢⵁⵢⵓⵢ
yaq	ⵢⵓ	q	ق	ⵢⵓⵢⵓⵢ
yi	ⵢⵉ	i	ي	ⵢⵉⵢⵓⵢ
yaj	ⵢⵓ	j	ج	ⵢⵓⵢⵓⵢ
yal	ⵢⵏ	l	ل	ⵢⵏⵓⵢ
yam	ⵢⵏ	m	م	ⵢⵏⵓⵢ
yan	ⵢⵏ	n	ن	ⵢⵏⵓⵢ
yu	ⵢⵓ	u	و	ⵢⵓⵢⵓⵢ
yar	ⵢⵓ	r	ر	ⵢⵓⵢⵓⵢ
yaɾ	ⵢⵓ	ɾ	ر	ⵢⵓⵢⵓⵢ
yay	ⵢⵓ	ɣ	غ	ⵢⵓⵢⵓⵢ
yas	ⵢⵓ	s	س	ⵢⵓⵢⵓⵢ
yaʂ	ⵢⵓ	ʂ	ص	ⵢⵓⵢⵓⵢ
yac	ⵢⵓ	c	ش	ⵢⵓⵢⵓⵢ
yaɫ	ⵢⵓ	t	ت	ⵢⵓⵢⵓⵢ
yaw	ⵢⵓ	w	و	ⵢⵓⵢⵓⵢ
yay	ⵢⵓ	y	ي	ⵢⵓⵢⵓⵢ
yaz	ⵢⵓ	z	ز	ⵢⵓⵢⵓⵢ
yaz	ⵢⵓ	z	ز	ⵢⵓⵢⵓⵢ

Figure 8: Official symbol equivalence from IRCAM [5]

When one attempts to look into how standardized **zgh** is, the initial impression is that clear standards exist due to a well-standardized writing system that has direct transliterations across the 3 writing systems (Figure 8). However, this is not so clear when looking at the phonological spaces of the languages that supposedly share this standardized writing (Tables 1,2,3,4). This is due to IRCAM having provided an official transliteration across the different writing systems and proposing a phonological transliteration for each letter in the standard (Figure 9) but different dialects adopting different realizations (and some spirantizations or assimilations undocumented by IRCAM).

Lieu d'articulation Mode d'articulation			Labiales	Dentales	Alvéolaires	Palatales	Vélaires	Labiovélares	Uvulaires	Pharyngales	Laryngale
Occlusives	Non emphatiques	Sourdes		+			ʔ	ʔ	ʔ		
		Sonores	ʔ	ʔ		ʔ	ʔ				
	emphatiques	Sourdes		ʔ							
		Sonores		ʔ							
Constrictives	Non emphatiques	Sourdes	ʔ		ʔ	ʔ			ʔ	ʔ	ʔ
		Sonores			ʔ	ʔ			ʔ	ʔ	
	emphatiques	Sourdes			ʔ						
		Sonores			ʔ						
Nasales			ʔ	ʔ							
Vibrantes	Non emphatiques			ʔ							
	emphatiques			ʔ							
Latérale				ʔ							
Semi-consonnes			ʔ			ʔ					

Figure 9: Official Tifinagh to Phonetics from IRCAM [5]

Consonants **rif**

	Labial - ɿ	Dental - ɿ	Alv. - ɿ	P. Alv. - ɿ	Palatal	Velar - w	Uvular	Phar.	Glottal
Nasal	m		n			ŋ			
Plosive vl.	p p ^ɿ		t t ^ɿ	tʃ		k k: ^w	q		ʔ
Plosive vd.	b		d d ^ɿ	dʒ		g g: ^w			
Fricative vl.	f	θ	s s ^ɿ	ʃ ʃ ^ɿ	ç	x ~ χ		ħ	
Fricative vd.	β	ð ð ^ɿ	z z ^ɿ	ʒ	(j)	ɣ ~ ʁ		ʕ	h
Approximant			l l ^ɿ		j	w			
Flap			r r ^ɿ						
Trill			r r ^ɿ						

Table 1: Consonants of **rif** according to [20]. "ɿ" as a superindex stands for pharyngealization. "w" as a superindex stands for labialization.

Consonants **shi**

	Labial	Dental - ɿ	P.Alv.	Velar - w	Uvular - w	Pharyngeal	Glottal
Nasal	m	n					
Plosive vl.		t t ^ɿ		k k ^w	q q ^w		
Plosive vd.	b	d d ^ɿ		g g ^w			
Fricative vl.	f	s s ^ɿ	ʃ		χ χ ^w	ħ	
Fricative vd.		z z ^ɿ	ʒ		ʁ ʁ ^w	ʕ	h
Approximant		l l ^ɿ	j	w			
Trill		r r ^ɿ					

Table 2: Consonants of **shi** according to [21]. "ɿ" as a superindex stands for pharyngealization. "w" as a superindex stands for labialization.

Consonants **tzm**

	Labial	Alv. - ɿ	Palatal	Velar - w	Uvular - w	Phar.	Glottal
Nasal	m	n n ^ɿ					
Stop vl.		t ^h t ^ɿ		k	q q ^w		
Stop vd.	b	d d ^ɿ		g			
Fricative vl.	f	s s ^ɿ	ʃ	(x) x ^w	χ (χ ^w)	ħ	h
Fricative vd.		z z ^ɿ	ʒ	(ʁ) ʁ ^w	ʁ (ʁ ^w)	ʕ	
Approximant		l l ^ɿ	j	w			
Trill		r r ^ɿ					

Table 3: Phonological space of **tzm** according to [22]. "ɿ" as a superindex stands for pharyngealization. "w" as a superindex stands for labialization.

Kabyle consonant phonemes

	Labial - w	Dental - ɿ	Alv. - ɿ	P.Alv. - ɿ	Palatal - w	Velar - w	Uvular - w	Phar.	Glottal
Plosive/Affr. vl.		t t ^ɿ	ts	tʃ		k k ^w	q q ^w		
Plosive/Affr. vd.	b b ^w	d	dz	dʒ		g g ^w	ɢ		
Fricative vl.	f	θ	s s ^ɿ	ʃ ʃ ^ɿ	ç ç ^w		χ χ ^w	ħ	h
Fricative vd.	β	ð ð ^ɿ	z z ^ɿ	ʒ ʒ ^ɿ	ɟ ɟ ^w		ʁ ʁ ^w	ʕ	
Nasal	m	n							
Trill		r r ^ɿ							
Approximant		l l ^ɿ			j	w			

Table 4: Phonological space of **kab** according to [23]. "ɿ" as a superindex stands for pharyngealization. "w" as a superindex stands for labialization.

Furthermore, when looking at the official script ²⁴ several letters are normally unused, and some IPA consonants that do not appear in IRCAM's standard appear due to spirantization. Not only that, the contrastive geminates from **shi** referenced by [9] are not mentioned in the Phonology of either (and are not always realized as long consonants in other Berber variants). Cross-referencing the phonology tables and listening to the datasets was necessary to build a solid **zgh orthography -> IPA dictionary**.

If one were to add **rif** to the languages handled, we come across issues particularly regarding making the Velar-Uvular fricatives. In the case of **rif**, these vary in the dialectal spectrum; where as in **tzm** Ayt ayache, /x/ and /ɣ/ are the pronunciation of "k" and "g" respectively (so we have some phonology collisions). "r" also becomes vocalized in **rif** following a vowel, and the stop -> fricative transformation that is common in **kab** also exists for **rif**. Given all these differences, the lack of corpus for it, and despite **zgh** being a standard form that includes **rif**, I have completely excluded it from my orthographic transliteration.

In my project, I have decided to create two separate transliterations for **kab** and **zgh** respectively; this seemed like the split that best represented the differences. If **rif** were ever to be added; probably using the **kab** transliteration would be more advisable.

4.1 Commonalities in transliteration

Generally speaking standardization is applied first and specific realizations or geminates are handled afterwards during transliteration to IPA. This transliteration simply matches a longer set of symbols when possible (to capture geminates or emphatic consonants). The pronunciation dictionaries resulting provide as many realistic realizations of a word possible and must be merged later; I will cover the particularities that this merge implies later when talking about the MFA specifically. I have tried to be as rigorous as possible with the IPA representations, but some unconventional choices exist among my character-to-IPA transliterations:

- **"e" may be realized or not** Listening to audios of both languages, it was clear that "e" is not necessarily always pronounced in both **kab/zgh**
- **Pharyngeal fricatives are transliterated as /ħ/, /ʕ/ and the glottal fricative as /h/**. In some of my sources pharyngeal fricatives are suggested to be epiglottals rather than pharyngeal; on the other hand, the glottal fricative is in some dialects voiceless (**tzm**) and in others voiced (**rif/shi**). My IPA choice among both epiglottal/pharyngeal and voicedness is arbitrary: I am confident that both may vary according to consonant context and no collision exists where any of the realizations of these consonants (or their geminates) would be mistaken for another by a native speaker.
- **IPA /j/, /tʃ/, /ʒ/, /dʒ/ are written in latinscript "c", "č", "j", "ğ" respectively**. I decided on this latinscript symbols because they seemed the least ambiguous on pronunciation for a native speaker.
- **Geminates are realized as long consonants (e.g. "tt" -> /t:/) excluding the exceptions mentioned in kab transliteration.**

²⁴https://en.wikipedia.org/wiki/Tifinagh#Neo-Tifinagh_letters

With regards to the rest of transliteration Tifinagh/Latinscript -> IPA, there are not that many remarkable choices; the standard Neo-Tifinagh symbols have well-known IPA pronunciations which have a non-ambiguous Latin equivalent. The Latin orthography for both follows the "Agreed" orthography (writing "y" as the pronunciation for "y"/"ɣ" which refers to IPA /j/ and "j" as the pronunciation for "j"/"ɟ" which refers to the IPA /ʒ/).

4.2 kab Transliteration

Implementation is available in the repo ²⁵.

kab, is a language primarily written in Latin script (and exceptionally in Tifinagh). This greatly simplifies standarization; my standarization dictionary handles mostly unofficial yet common ways of writing a same character. Only a couple choices are worth mentioning with regards to standarization:

- **"tt" and "dt" become standarized to the character "ṭ"** Geminate "t" in **kab** is pronounced /ts/; while the symbol "ṭ" is not used anymore; it used to be a way of writing the sound /ts/ across Berber variants. This could be handled when I transliterate to IPA, but since the non-ambiguous symbol "ṭ" exists for it I found it better to use it and transform this symbol to /ts/. It makes the **symbol -> IPA dictionaries** more similar between **kab/zgh**. In **"dt"**, /d/ assimilates to /t/; making it effectively "tt" and pronounced /ts/.
- **"zz" becomes normalized to the character "ẓ"** Same reasoning as the geminate "t" applies here, but here we handle the voiced affricate /dz/.

The simplicity of the standarization is offset by how complex transliteration to IPA in Kabyle is. Roughly speaking, most consonants equivalent to stops in **kab** are pronounced by default as fricatives (in a predictable manner) and depending on the preceding context (or when they appear as geminates) become proper stops ²⁶. This is something I have contrasted and confirmed by talking to a Native speaker ²⁷. A series of assimilations are also listed in Wikipedia, which have been included according to whether the Native speaker had heard them at some point; the idea being that I want to have available for the aligner all the realistic pronunciations of a word. To handle this; 2-word transliteration entries have been used. The noteworthy choices are:

- **"b" is pronounced /β/** except when preceded by "m", as suggested by the sources.
- **"d" is pronounced /ð/** except when preceded by "l"/"n", as suggested by the sources.
- **"t" is pronounced /θ/** except when preceded by "l"/"m"/"n", as suggested by the sources.
- **"g" is pronounced /j/** except when preceded by "b"/"r"/"z"/"ɛ"/"j", as suggested by the sources.
- **"k" is pronounced /ç/** except when preceded by "f"/"b"/"s"/"l"/"r"/"n"/"h"/"ɛ"/"c", as suggested by the sources.

²⁵https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/kab_build_dicts.py

²⁶https://en.wikipedia.org/wiki/Kabyle_language#Fricatives_vs._stops

²⁷<https://www.youtube.com/@massinissaait-ahmed4875>

- "nw" may be pronounced /nw/, /npw/ or /nbw/ following the pronunciation of the native speaker.
- geminate "ww" may be pronounced /bː/ or /w:/ , as suggested by the sources.
- geminate "yy" may be pronounced /g:/ or /j:/ , as suggested by the sources.
- geminate "yy" may be pronounced /q:/ or /ɣ:/ , as suggested by the sources.

4.3 zgh Transliteration

Implementation is available in the repo ²⁸.

In the case of **zgh** my initial intention was to cover the 2 advisable writing systems (Tifinagh and Latinscript). Since the dataset in Latinscript²⁹, labelled as **zgh**, was discarded due to being **kab**; my standarization and transliteration for Latinscript are currently unused. Both still remain in the codebase and generate Latinscript entries in the pronunciation dictionary for words that were found in Tifinagh. They are left there in case someone were to further work on this project. Therefore, only Tifinagh transliteration to IPA will be described in this section.

To create a pronunciation dictionary that was as robust as possible for all the **zgh** variants, I cross-referenced many relatively reliable sources and chose which phone best represented the Tifinagh character and which translation is proper. My sources were:

- https://en.wikipedia.org/wiki/Tifinagh#Neo-Tifinag_letters
- <https://en.wiktionary.org/wiki/Module:Tfng-translit>
- <https://www.mdpi.com/2078-2489/16/7/600>
- <https://ieeexplore.ieee.org/abstract/document/8284715>
- <https://commons.wikimedia.org/wiki/Tifinagh>
- https://en.wikipedia.org/wiki/Berber_Latin_alphabet
- The Phonology pages for the 3 relevant variants
 - https://en.wikipedia.org/wiki/Shilha_language#Phonology
 - https://en.wikipedia.org/wiki/Central_Atlas_Tamazight#Phonology
 - <https://en.wikipedia.org/wiki/Tarifit#Phonology>

Some of the characters that appear in these pages are not part of the Berber standard alphabet; but my aim was for the transliteration to cover them. Standarization shows my choices for dealing with them:

²⁸https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/zgh_build_dicts.py

²⁹https://huggingface.co/datasets/TutlaytAI/moroccan_amazigh_asr

- " $\oplus$ ", " $\times$ ", " $\vee$ ", " $\exists$ ", " $\wedge$ ", " $\exists$ " are normalized respectively " Θ ", " $+$ ", " $\wedge$ ", " E ", " $\wedge$ ", " $\exists$ ".

All these are part of the extended Tifinagh letters, that cover a spirantization of the base symbol (thus their resemblance). As far as I understood speaking with a native speaker, these are mostly used when the pronunciation of a word in a specific dialect wants to be emphasized. I normalize these, because I am unsure of whether a native speaker will respect/imitate another's dialect pronunciation. Spirantizations should also be properly understood as realizations of the base characters by our Aligner.

- "Ƶ", "ⵓ", "ⵓ:", "ⵓ:", "ⵓ:", "ⵓ:", "ⵓ:", "ⵓ:" are symbols belonging to other Berber languages. Most of these come from wiktionary; which is a decently reputed source for languages. The issue with this resource had mostly to do with how the resource was not using standard IPA; but cross-validating with other sources I used whichever pronunciation was the most common. With the symbols "Δ", "Ɔ", "ⵓ:" (supposedly "v", "p", "χ") I found no pronunciation examples and the symbols are not standard **zgh**. Being unable to confirm their pronunciation I decided to leave them commented. For my final transliteration version, I would rather only leave symbols where I have a good idea how they're pronounced (e.g. "v" could be reasonably pronounced /v/, /b/, /β/ or /f/; so I exclude any sentences containing it).
- **Affricate non-Neo-Tifinagh symbols "ⵓ", "ⵓ", "ⵓ", become their respective multisymbol representation "ⵓ", "ⵓ", "ⵓ"** This is purely to stick to a more standard writing for the Tifinagh pronunciation dictionary. I expect in natural writing for both to appear.
- **"ⵓ", "ⵓ" which should be /ny/ and /ng/ respectively** Were referenced by both wiktionary and wikimedia, however I found no pronunciation examples once more; and in this case I would suspect them being the /n/ and /ŋ/ respectively. They have been left as code comments.

With regards to the rest of transliteration Tifinagh -> IPA, there are not that many remarkable choices:

- **Gemination is transcribed as the long version of its IPA consonant.** Unlike in Kabyl, where there are some irregularities with gemination.
- **Labialized emphatic consonants have a special writing** E.g. "𐵓𐵓" is the standard writing, not "𐵓𐵓". I saw no documentation of this being mentioned, but detected it during manual labeling, and it has been left corrected in the code.

5 MFA: Implementation details

The project pipeline is best described by looking at the **README.md** from the repo ³⁰. One has to create a clean environment with conda (as suggested by the MFA documentation ³¹), then install whichever packages are intended to use (I would usually use requirements.txt; but in this case there's not enough libraries to warrant it). The setup for multithreading is completely optional, but extremely advisable. The strategy for generating a good **shi** aligner here is to first train an MFA on **kab**, then adjust it to the shorter audios from **zgh**, and use the longer audios as separate testing for **tzm/shi**. The **README.md** then goes over the different corpus setup steps:

- **prepare_paths.py** Which cleans preexisting files from all the folders relevant to this project. This is the only script that should be run before setting up the rest of the corpus; the rest can be run in any order.
- **{kab|zgh}_build_dicts.py** and **merge_dicts.py** Which generate the word dictionaries for each respective language (**zhg** including **shi** and **tzm**) given the previously mentioned orthography (and include in-code the source for each "symbol->phone" pair) that will be used when training the MFA specific model. When merging the dicts, 2 versions of the joined dictionaries are made: one that maximizes the probabilities for **zgh** and minimizes the probabilities for **kab** and one that does the reverse. The intention here is to prevent the model from mixing up languages when training/adjusting. Note also that while building the dictionaries we will be using the testing samples. This is not ideal, but testing without adding them resulted in a very high OOV ratio, that made comparison to manually labeled examples useless. In the interest of evaluating the MFA performance and given the limitations faced when finding reliable data for **zgh**, this is a concession I had to make. All in-word assimilations are handled here. **n**, **ad** and **rad** have unique assimilations according to [9] and these assimilations have been implemented in **zgh_build_dicts.py**; other assimilations from [9] are also cited as comments with their respective page, even though are left unused.
- **{kab|zgh}_gen_corpus_acoustic_model.py** These two scripts generate the proper training corpus that will be used to train/adjust or MFAs. The testing samples on **shi/tzm** are excluded from **zgh**.
- **{shi|tzm}_gen_corpus_acoustic_model.py** These two generate the testing corpus that will be compared with the manually/golden labeled data.

As is advised by the MFA documentation ³² I resampled all the audio to the "lingua franca" of audio formats (16kHz wav + 16bit precision) to significantly lower the resources required for future training (implemented in the **X_gen_corpus_acoustic_model.py** scripts). Audio/text preprocessing has also been adjusted to be able to work in streaming mode, meaning that a computer with under 20GB of RAM should be able to process the totality of data. Multithreading setup was also important, using 12 cores I can generate the biggest corpus (**kab**) in under 30 minutes.

³⁰under "FULL run for dissertation/reproduceability" https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/README.md

³¹https://montreal-forced-aligner.readthedocs.io/en/stable/getting_started.html

³²https://montreal-forced-aligner.readthedocs.io/en/latest/user_guide/troubleshooting.html#speed-optimizations

Further along the **README.md**, the only steps that are left are to train and evaluate the MFA models themselves. MFA allows on-training specifying several speakers and adapting the model to listen to each of them; which have the limitation of being single-threaded per-speaker, but may be helpful in a meeting. This aspect has been ignored as the interest of the thesis is to make a generalistic model; the parameter **-single-speaker** is always used to make the MFA consider each audio a separate speaker. MFA has also two ways to deal with multiple realizations of a single phone:

- **Phonological Rules**³³ Which are simple regular expressions that replace/remove/insert character given specific in-word examples, and generate for every pronunciation a series of variants. These are applied on the pronunciation, rather than the writing of the word. I found it clean to use replacement rules to handle cross-word assimilations, as mentioned in-word assimilations are already handled when using the orthography to generate the pronunciation dictionaries. As I found conflicting sources (listed in https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/rules/zgh.yaml) for different spirantizations/assimilations; I found it wiser to limit myself to the ones I have seen widely agreed upon. I only wrote geminate cross-word assimilation and the relevant to apply the No-Hiat rules; which are also the ones I saw commonly during Manual evaluation.
- **Phone Groups**³⁴ Even though I wrote two versions; I have limited myself to analyzing the more intuitive version that can be found in https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/phone_groups/kab_ortho.yaml. Here all similarly pronounced phones are grouped, meaning that the **kab** fricatives belong the same group to their plosive counterpart. Simply put, I think this should not be an issue as these do not exist in **zgh** and I do not trust the contexts where these fricatives turn to their plosive counterpart to be so set-in-stone. Emphatic consonants have also been left in the same group. My alternative suggestion (in https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/phone_groups/zgh_ortho.yaml) fine-grains a bit more separating emphatic consonants to their simple version; the idea here being since these are lexically contrastive they may be worth separating phonologically as well. The phone groups are applied in the triphone step of the MFA and basically set up the same decision tree for phones in the same phone group.

After running the training/adjust commands and aligning the particular test samples, a simple script exists to generate all the output plotting and csvs under the **plots** folder. Once again for the sake of reproducibility/reporting these are available in the repository ³⁵. As can be seen, all plots and alignments are generated for both training with **kab** and training solely using the limited **zgh** corpus.

³³https://montreal-forced-aligner.readthedocs.io/en/latest/user_guide/implementations/phonological_rules.html

³⁴https://montreal-forced-aligner.readthedocs.io/en/latest/user_guide/implementations/phone_groups.html

³⁵https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/plots/

6 Analysis conditions and reproducibility

To select the most representative samples, the manual annotation was done on the longest (by text) first 20 samples for both **shi** and **tzm**. The remaining of the **zgh** corpus has been used for adjusting (excluding the 40 testing samples); and a random set of 40k **kab** samples were chosen with appropriate seeding³⁶. After some filtering the **kab** corpus had a total of 88,006 seconds (24.45 hrs) and **zgh** 1,967 seconds (32.7 min).

For the sake of grading this dissertation and future reproducibility I have uploaded to the repository holding this project my manual labeling, all the Textgrids from training and adjusting, all the generated plots and naturally all the code involved in this project. To give a general idea of the relevant folders and files:

- **Automatically labeled data by MFA using kabyl as pretraining** https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/output_kab
- **Automatically labeled data by MFA without using kabyl as pretraining** https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/output_nokab
- **Manually labeled data** https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/output_verified
- **General folder for graph visualizations** https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/plots
- **Common voice dataset sentences** https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/raw_sentences
- **Script run to generate plots** https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/evaluate_alignments.py
- **Private dataset plots** https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/plots/secret Plots using testing data privately shared by Prof. Gafos. I cannot release the data publically, but general plots are fine.

Last commit upon this delivery was: 49dcf724a0c18b4edb7f36f9dd03e6e405a1468a. Code-base is available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/49dcf724a0c18b4edb7f36f9dd03e6e405a1468a

³⁶see function `load_datasets_kab` in https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/utils.py

7 Histogram analysis

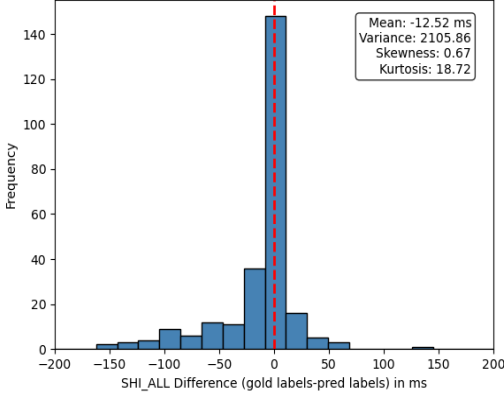
7.1 Comparison on common voice: "Training on kab and adjusting to zgh" vs "Training with zgh"

Looking at the histograms in Figures 10 and 11 it is very clear that using **kab** as initial training results in much smaller errors for both dialects (more results in the buckets near-0), both the variance and the mean are closer to 0 for every model that was trained using **kab**. This concurs with other research [10], we can see in Tosolini's paper that the most relevant for labeling with near-0 difference was having seen a familiar language. The sample number in our case is much lower; resulting in a more accidented distribution. In particular we can see that **tzm** becomes dramatically worse without the **kab** corpus. There are 2 reasons for this:

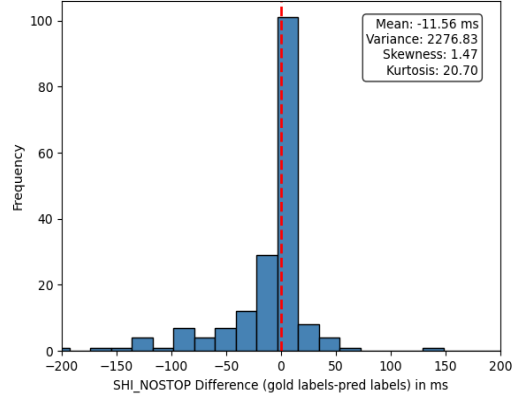
- **tzm** has a tiny amount of samples when excluding testing (less than 40) so the aligner is very **shi**-centric. The models for different phones may be very fine-tuned to what is expected of a **shi** speaker and not so generalistic.
- **tzm** is part of a dialect continuum **shi-tzm-rif**. **rif** is also well-known to share a lot similarities with **kab**, so when we are training using **kab** we are not just showing further possible realizations in a berber language, rather the other side of the spectrum of realizations that **shi** does not represent.

Please note that I am using per-word calculated overlap difference here, in comparison to Tosolini's [10] boundary onset.

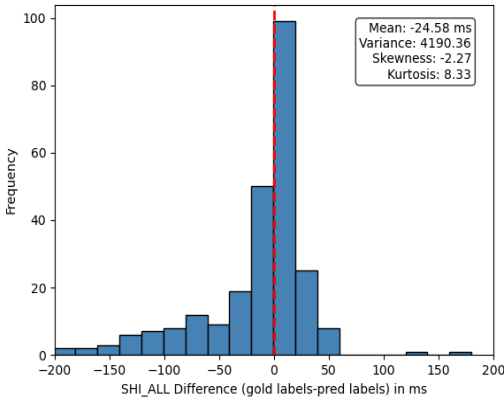
Comparing the **kab**-pretrained histograms between **shi** and **tzm**, one may think at first that MFA performs better on **tzm** since the 0-peak is higher; however the variance is higher in **tzm** for both criterias of evaluation (including 1-2 length words or not). This implies that the aligner performs more reliably on **shi**, so it probably generalized better for it. The skewness flips between **shi** and **tzm**, which probably means our aligner did not generalize enough overall and would benefit from more corpus on both ends. I attribute the extremely high kurtosis to the low sample size.



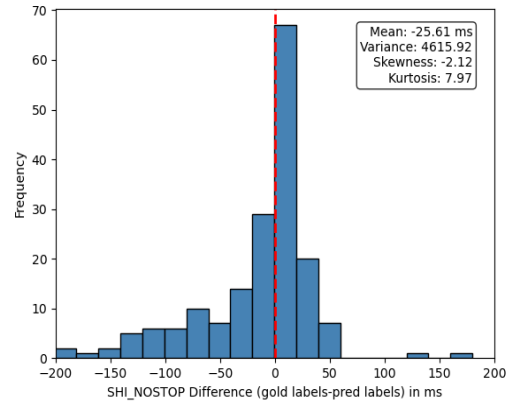
(a) Differences using **kab** as pretraining, evaluating on all words.



(b) Differences using **kab** as pretraining, evaluating on all words except those with length 1 or 2.

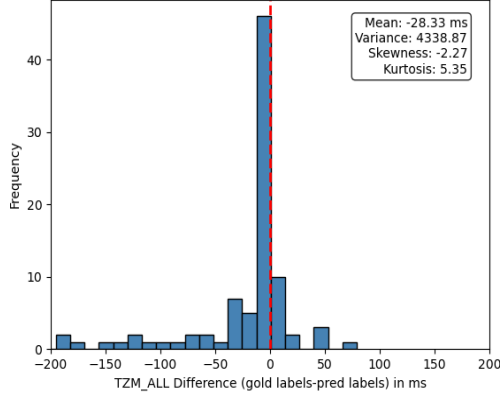


(c) Differences without using **kab** as pre-training, evaluating on all words.

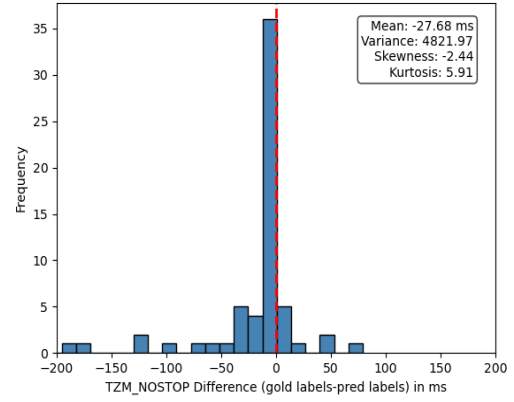


(d) Differences without using **kab** as pre-training, evaluating on all words except those with length 1 or 2.

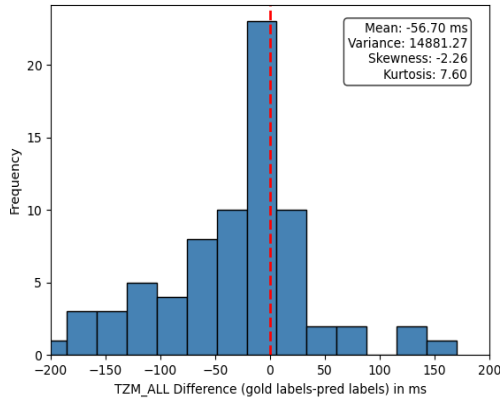
Figure 10: Histogram of word differences (gold-predicted) in miliseconds across the **SHI** aligned dataset. Available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/plots/output_nokab and https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/plots/output_kab



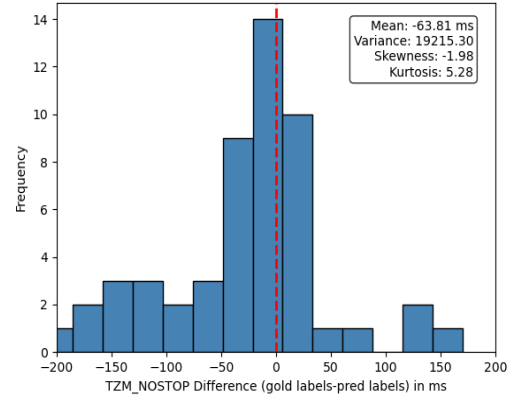
(a) Differences using **kab** as pretraining, evaluating on all words.



(b) Differences using **kab** as pretraining, evaluating on all words except those with length 1 or 2.



(c) Differences without using **kab** as pre-training, evaluating on all words.



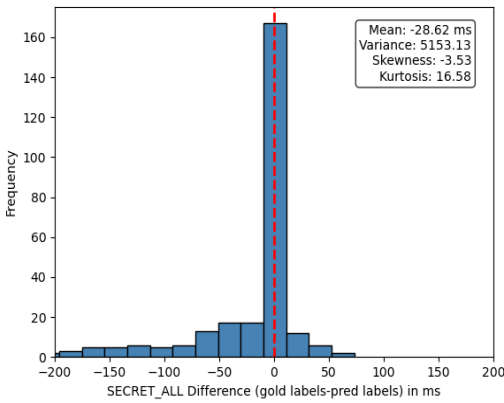
(d) Differences without using **kab** as pre-training, evaluating on all words except those with length 1 or 2.

Figure 11: Histogram of word differences (gold-predicted) in miliseconds across the **TZM** aligned dataset. Available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/plots/output_nokab and https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/plots/output_kab

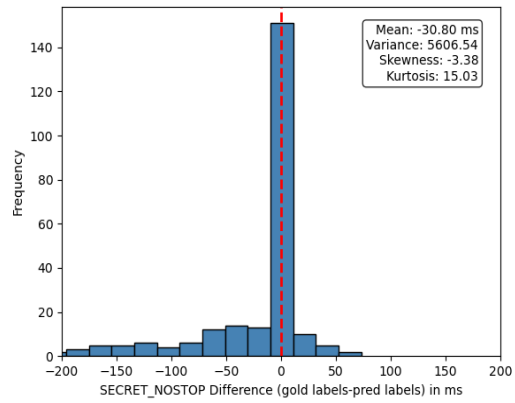
From this subsection going forward, the plots used are based on the MFA that was pretrained in **kab**, since it performs predominantly better.

7.2 Private data

We can notice the model labels very skewed towards the left: the beginning of the word tends to be labeled later than it should (the model tends to label the first phone late, missing the first character). The private data contained **shi**, but it seems to perform closer to **tzm**. This is probably due to the private data containing single-file multi-sentence audios; a big error on labeling a word at any point will carry over for the following words.



(a) Differences using **kab** as pretraining, evaluating on all words.



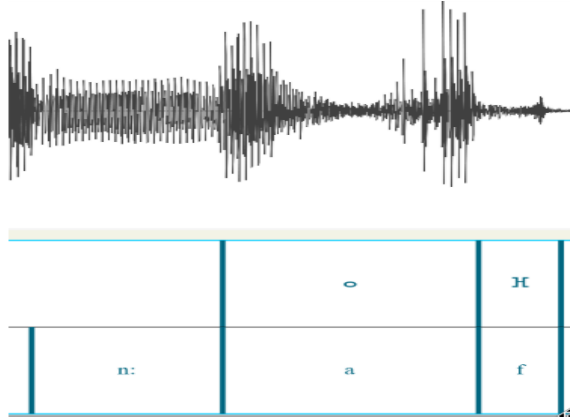
(b) Differences using **kab** as pretraining, evaluating on all words except those with length 1 or 2.

Figure 12: Histograms of word differences (gold-predicted) in milliseconds across the private aligned dataset.

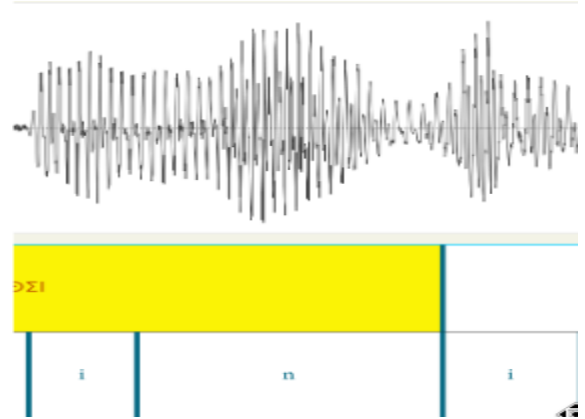
8 Manual analysis

8.1 Common Voice

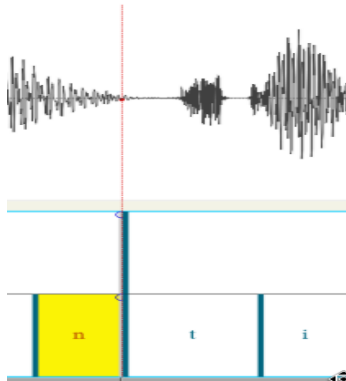
8.1.1 Dealing with audio irregularities



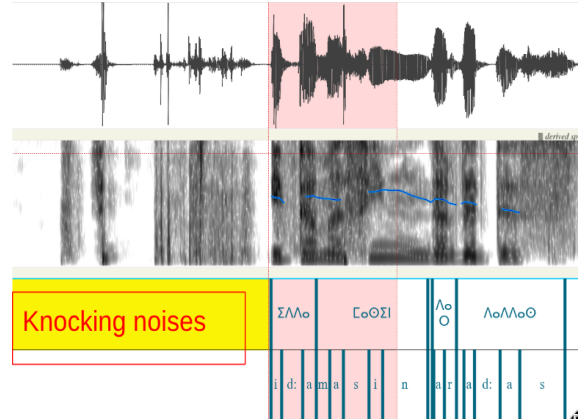
(a) Automated MFA labeling. MFA correctly labeling hiccups within a vowel (without detecting them as a different phone). Audio sounds like /ehə/ *shi/common_voice_22_O_0*



(b) Automated MFA labeling. MFA correctly labeling vocoids adjacent to /n/ (without detecting them as a different phone). Audio sounds like /nən/ *shi/common_voice_22_O_3*



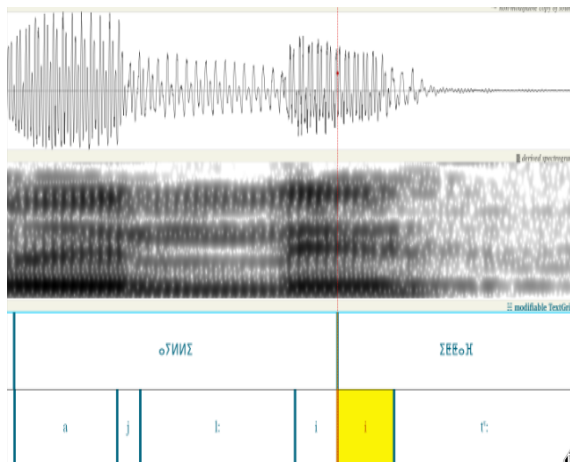
(c) Automated MFA labeling. MFA correctly labeling hiccups within a partially released /t/ (without detecting them as a different phone). *shi/common_voice_22_O_1*



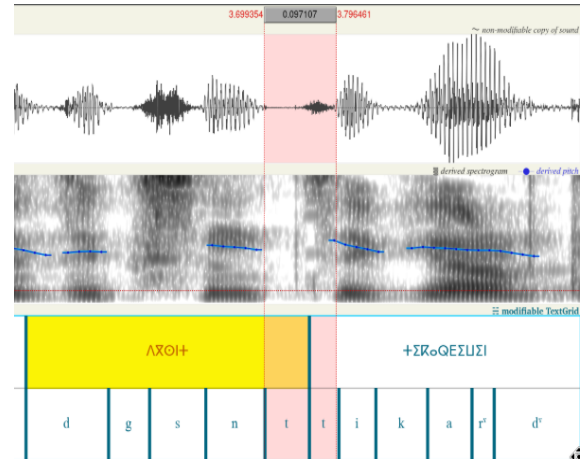
(d) Automated MFA labeling. MFA correctly ignoring knocking noises. *tzm/common_voice_22_O_17*

Figure 13: Examples of the MFA correctly labeling despite in-phone irregularities. These are all from MFA's automated labelling, and an example has been left in the repo for the sake of grading this dissertation at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/output_kabylpretraining/

Figure 13 shows us how the MFA is able to successfully label despite heavy in-phone irregularities.



(a) No-Hiat example from [9] pages 190-193 being ignored. One of the /i/ should become /j/. *shi/common_voice_22_o_2*

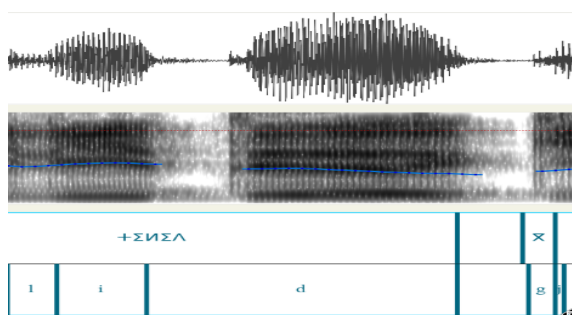


(b) According to [9] two individual /t/ at the beginning and end of a word will merge into a geminate /t:/. This is corroborated by looking at the wave. *shi/common_voice_22_o_5*

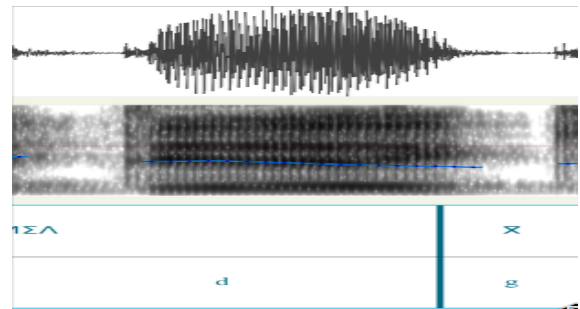
Figure 14: Automated labeling MFA. Examples of crossword interactions not being handled properly. While the rules required for the proper replacement are specified in https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/rules/zgh.yaml, they have not been used by the MFA.

Figure 14 shows us how the MFA does not perform well, in 2 assimilations that are referenced by the main source used for *shi* orthographical transcription (and that happen as described by the reference). The rules exist for these to be properly labeled, but I presume the aligner did not see enough corpus to use those alternative spellings.

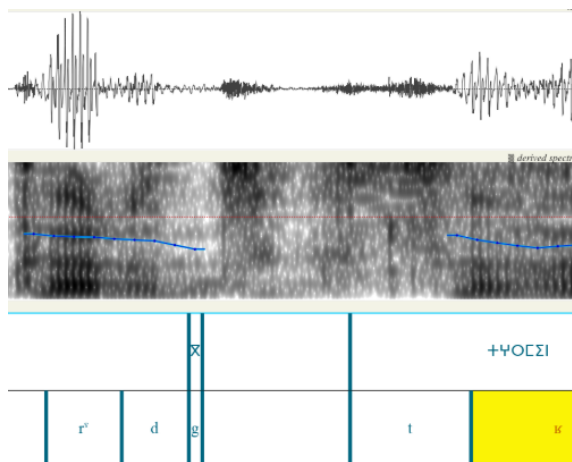
8.1.2 Single-consonant preposition mislabels and assimilations



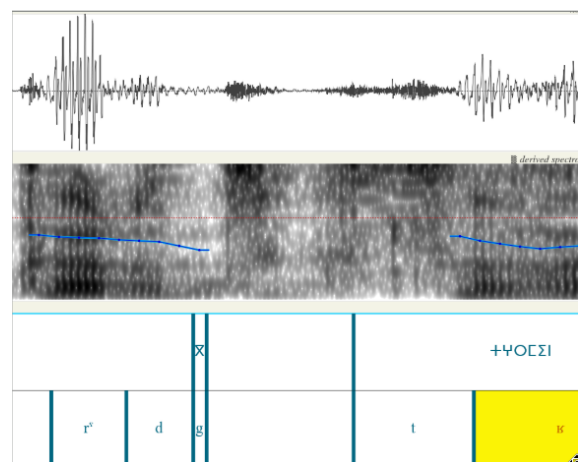
(a) Automated MFA /g/ mislabeling. *shi/common_voice_22_O_4*



(b) Manual corrected /g/ labeling. /d/ includes the vocoid after it. *shi/common_voice_22_O_4*



(c) Automated MFA /g/ mislabeling. *shi/common_voice_22_O_4*

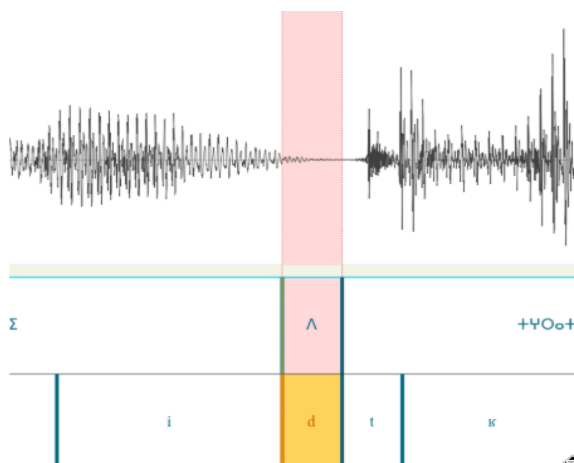


(d) Automated MFA /g/ mislabeling. *shi/common_voice_22_O_5*

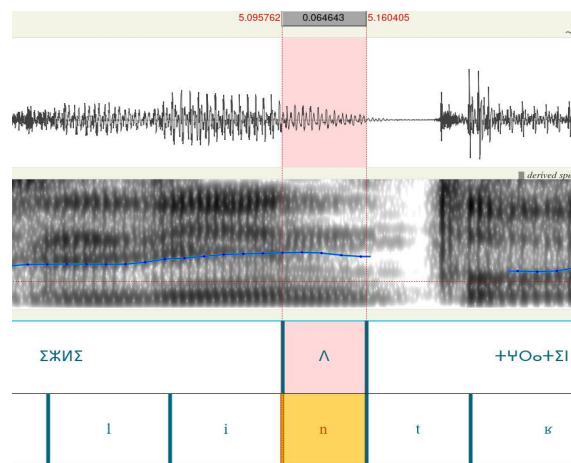
Figure 15: Examples of single-consonants mislabeling. (a) and (b) are the same segment. Two things must be remarked here: vocoids are manually labeled within the preceding consonant and MFA struggles with adding boundaries to single-consonant words. The second is specially true when in the vicinity of a vocoid.

Figure 15 shows how the MFA has a tendency to handle plosive by labeling only the air expulsion, excluding the descent to silence and silence happening during air retention (this also happens in /t/). As the uni courses have shown me, I have decided to label the entire segment as a plosive.

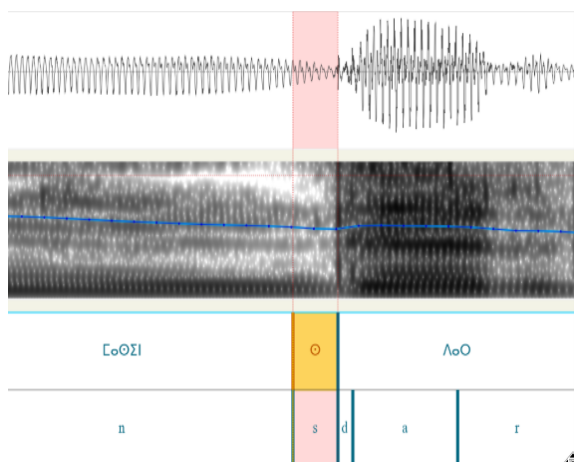
Figures 15 (a)(b) show another critical decision: how vocoids are handled. I have decided to handle them as part of one of the surrounding consonants. This is to be fair to the MFA: in my orthography the models have no way of knowing these vocoid phones are there. An alternative choice (that I would endorse for research purposes) would be to add for every possible articulation place between consonants that can have a vocoid an optional vocoid (which would make the amount of entries in the pronunciation dictionary explode). This would require an alternative manual labeling.



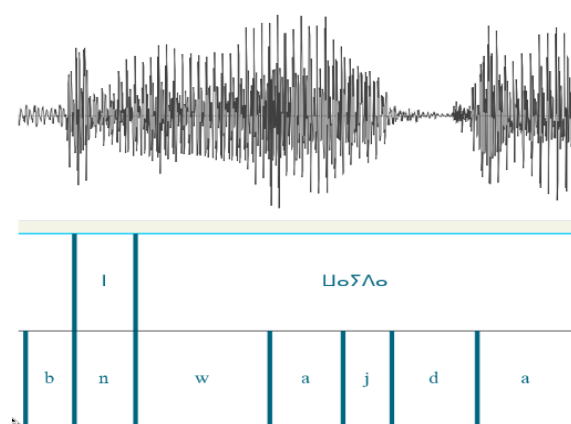
(a) Automated labeling MFA incorrectly labels as /d/. *shi/common_voice_22_o_2*



(b) Manual Labeling of corrected /n/. *shi/common_voice_22_o_2*



(c) Automated MFA Labeling. "s" preposition becomes completely assimilated. *tzm/common_voice_22_o_12*



(d) Manual labeling. According to literature [9] (page 46), /n/ preposition should assimilate to an upcoming /w/ by being realized as /u/; but does not do so and remains as /n/. *shi/common_voice_22_o_5*

Figure 16: Set of examples showing unusual assimilations for single-consonant-words. (a)(b) show the irregular assimilation of /d/ as /n/ prior to a /t/.

Looking at Figure 16 (d) "n" preposition should (according to [9] page 46) assimilate in this case, but it does not. Having gone through the other samples chosen for manual labelling, the literature suggestions for this particular preposition being assimilated does not feel true for the general **shi** or **tzm**. Looking at the other examples however, there is a pattern of single-consonant-prepositions being pronounced in unexpected ways based on the orthography. Perhaps their correct strict pronunciation is not so critical for phonetically parsing a sentence.

8.1.3 Geminate underperformance

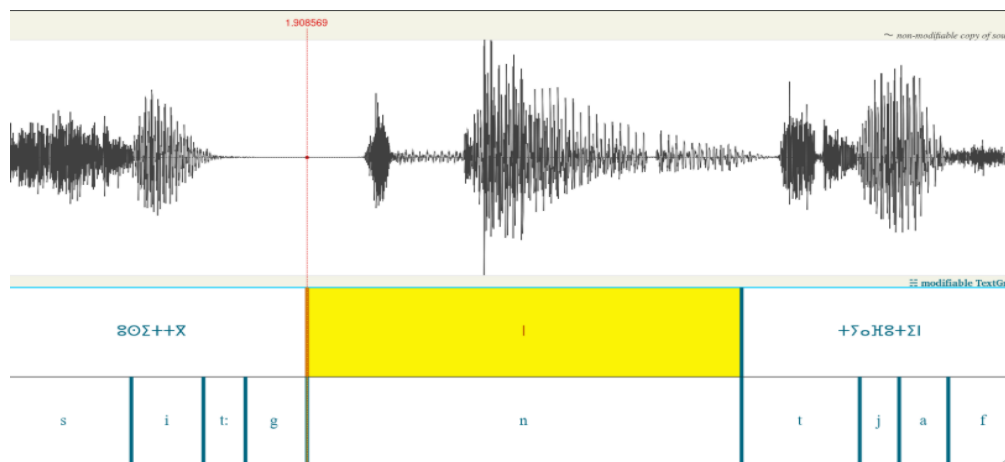
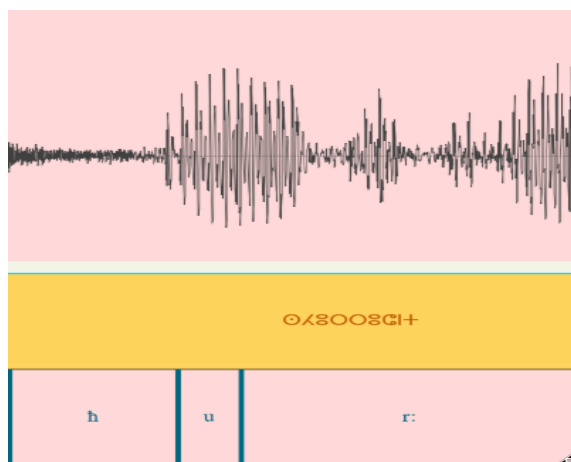
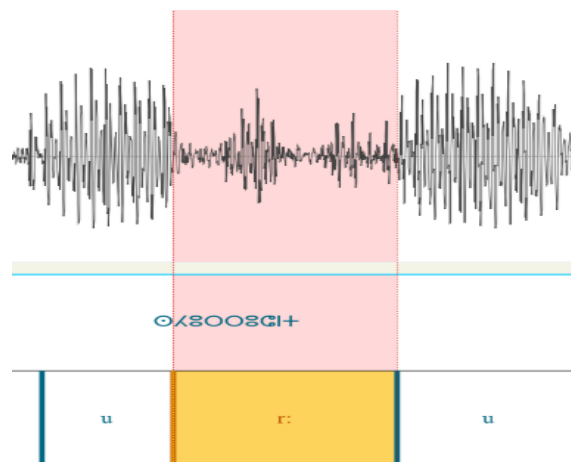


Figure 17: Automated MFA labeling. Surroundings of geminate t are completely mislabeled. *shi/common_voice_22_O_1*

The MFA clearly underperforms when a geminate t occurs. Figure 17 shows an example where the entire environment gets mislabeled due to a /t:/. This makes a lot of sense, since most of the training corpus is **kab** and no orthography exists of geminate t, "tt" is pronounced /ts/.



(a) Automated MFA labeling.

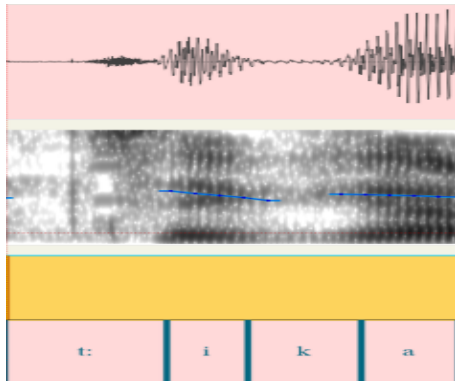


(b) Manual labeling.

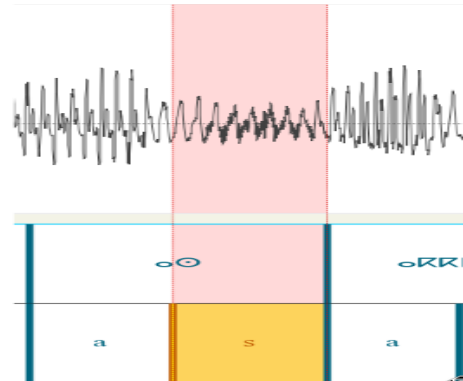
Figure 18: Geminate r mislabeling *shi/common_voice_22_O_2*

In Figure 18 it can be seen how the boundaries for rhotics and vowels are hard for the MFA, even when they are simple to distinguish in plain sight. There is a possibility that the issue lies with /r:/, it being rarer in the corpus than /r/. I can anecdotically say the same issue does not happen for does not happen for the glides /w/ /j/; and the nasals /m/ /n/.

8.1.4 Voicing

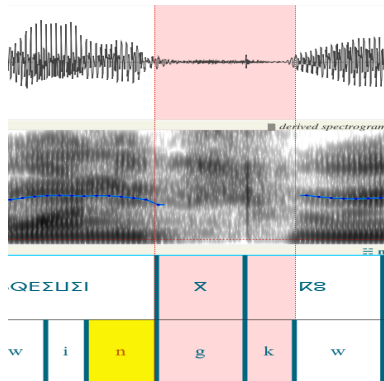


(a) Automated labeling MFA. /k/ becomes voiced /g/ (between vowels and voiced such as /n/ /r/). *shi/common_voice_22_o_5*

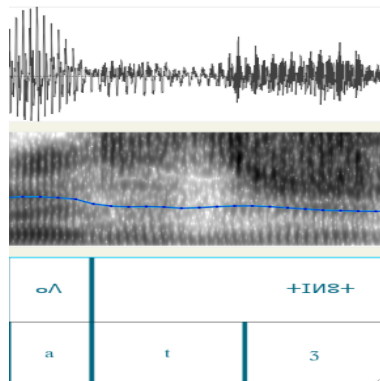


(b) Automated labeling MFA. Intervocalic /s/ becomes voiced /z/. *shi/common_voice_22_o_3*

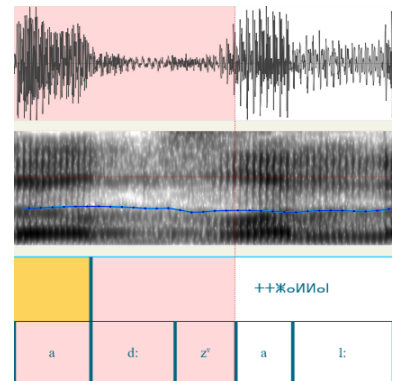
Figure 19: Simple Consonant voicing examples.



(a) Manual labeling. /g/ assimilates to /k/ AND spirantizes resulting in the highlighted segment being pronounced as /ç:/. Left as /gk/ to be fair to MFA. *shi/common_voice_22_o_5*



(b) Automated labeling MFA. /adtz/ which should be /at:z/ is realized as /az:z/. *shi/common_voice_22_o_6*

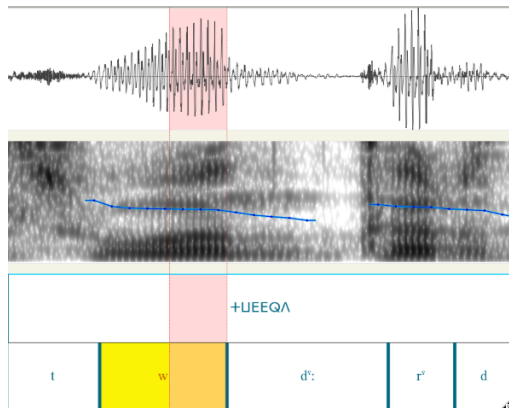


(c) Manual Labeling. "++" which should be /t:/ becomes voiced /d:/ . *tzm/common_voice_22_o_7*

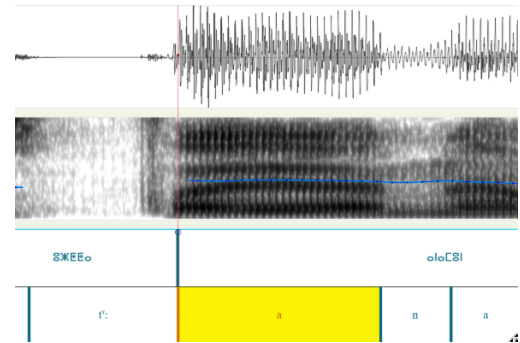
Figure 20: Geminate voicing.

Looking at the segments from Figures 19,20 it is clear that consonants become voiced when surrounded by voiced consonants and/or vowels. While this is mention in the literature [9], according to it voicing/devoicing should not propagate into geminates. In the examples given, this happens in both **shi** and **tzm**; the book specifically states how that does not happen in **shi**. It is possible that the lack of voicing for geminates is only excluded of the Imdlawn area (that the literature covers) or that the audios from common voice happened to only represent a specific geographic area; as a non-native I cannot determine which of those two may be true.

8.1.5 Epenthetic consonants



(a) Manual Labeling. An epenthetic /n/ can be heard in the highlighted segment. *shi/common_voice_22_O_5*

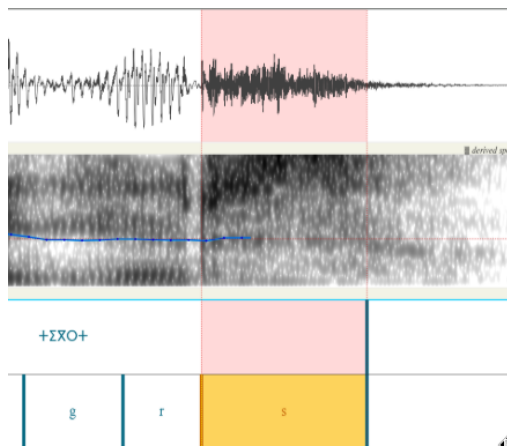


(b) Manual Labeling. An epenthetic /j/ can be heard in the middle of the highlighted /a/. *shi/common_voice_22_O_10*

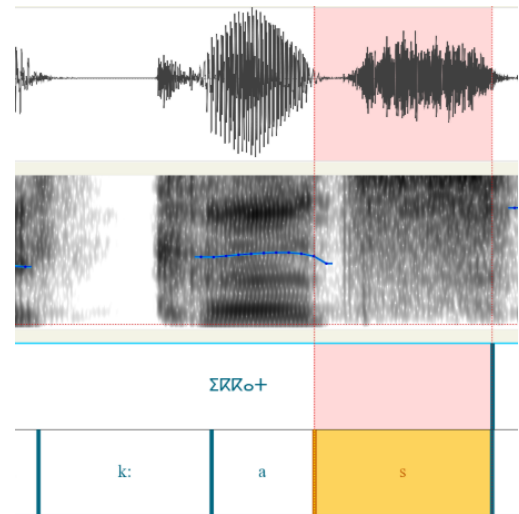
Figure 21: Epenthetic consonants can be heard clearly in some samples

I have also not seen such a phenomena being described; but Figure 21 shows us an example of epenthetic consonants being used during long vowels or glides. In the case of an /n/ happening during a /w/; it is probably to ease the pronunciation of the following pharyngealized d, but the second feels as irregular as a hiccup. I dare not say whether either of these are a pattern, but they are worth a mention.

8.1.6 Final-/t/ realizations



(a) Manual Labeling. "t" which should be /t/ is realized /s/. *shi/common_voice_22_O_9*



(b) Manual Labeling. "t" which should be /t/ becomes /ts/ before a pause. Perhaps **kab** influence. *tzm/common_voice_22_O_12*

Figure 22: Epenthetic consonants can be heard clearly in some samples

Something else worth a mention for is how final /t/ has unusual pronunciations. In Figure 22(a) we can see /t/ spirantizing to /s/ (other examples show /t/ at the same time spirantizing and voicing to be realized as /z/). I am uncertain of the conditions for this spirantizations to happen, but preceding a proper pause seems to enforce it. It is also noteworthy how it realizes as /ts/ in the **tzm** samples.

8.1.7 Emphatic consonants assimilate as normal consonants

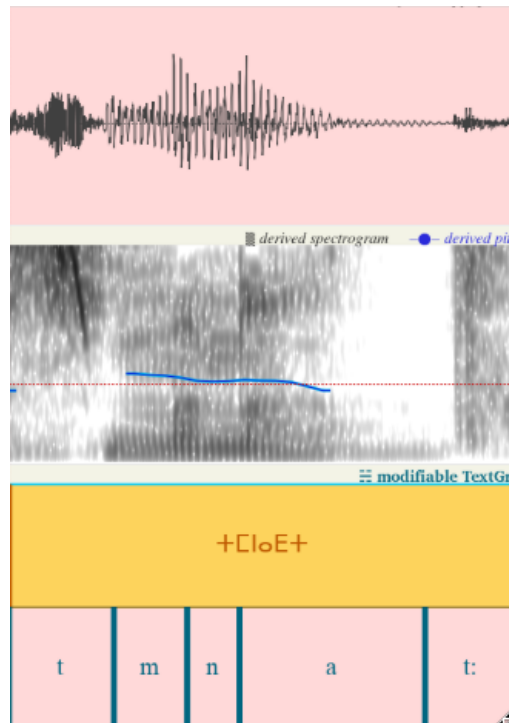


Figure 23: Manual Labeling. Pharyngealized d "E " assimilates to a following /t/ being realized as /t:/ (like a non-emphatic /d/). *shi/common_voice_22_O_14*

In Figure 23 we see an example of how a pharyngealized consonant assimilates as a normal one. While writing the assimilation rules I saw no comments regarding emphatic consonants, so I did not write this assimilation.

8.2 Private dataset

I will have no particular images to show from the private dataset, as I have been explicitly requested not to share the sentences. These were handpicked with a particular criteria not clear to me. As mentioned in the histogram sections, these are very long audios with a collection of sentences. I can say overall that the sentence order labelling performed by the MFA was acceptable (at no point a sentence was more than 1 sentence-distance away from where it was labelled).

The female speakers of these audios seem to speak some particular dialect where the spirantizations /k/ -> /ç/ and /g/ -> /j/ happen; just like in **kab**. These are evident in "kj" clusters, but can be verily commonly heard. As a personal limitation, I do not trust to hear the difference between /g/ and /j/, so please take my manual labeling with a grain of salt regarding these consonants. The MFA handles surprisingly well /f/ to /ç/ transitions (even though their wave looks very similar to the naked eye). Comparing the Common Voice corpus used for labeling; /k/ is a lot rarer there for both **tzm** and **shi** for Common Voice.

The audios themselves contain a very high amount of nasals surrounding consonants, which result in strange realizations. Internasal /t/ is not a plosive, rather a half-retenting of the /n/ phones surrounding it; it reminds a bit of the commonly used sound for affirming /nhn/. The MFA does surprisingly well (better than I) detecting and hearing these consonants, as well as several very short consonants that when listening to a single word I could not identify (/l/ or /t/ are sometimes very short). In general final /t/ (and to a minor degree /k/) are very hard for the MFA to properly label: this could be due to them being very quiet or how they tend to be very distant to their preceding phone in these audios .

In general I have noticed some other particularities regarding specific consonants:

- /j/ and /g^w/ are labeled by the MFA always shorter than they should.
- Sentence-final /n/ is very often missing its later half; even though between phones it gets properly labeled.
- The transition /j/ to /l/ is always very off.
- /h/ is usually mislabeled (even though it is common in kab corpus)
- Voids always surround /g/ (making it harder for me to identify when I should label /j/)
- In general, emphatic consonants labelling is very off by the MFA.

9 Statistical analysis: per-phone

9.1 Common Voice

To get a general idea of where the MFA fails more often, I decided to do a per-phone confusion matrix comparing the overlaps between the manually labeled alignments and MFA's. To ensure the matrix is showing only labelings with high support, I removed (set to 0) any metric based on under 8 occurrences, and after this removed rows/cols with only 0s. This would allow us to detect the phones where the MFA struggles with consistency. **UNK** represents the overlap with an unlabeled interval. In this analysis I will use more often confusion matrixes normalized by column because more information is visible. The entirety of this confusion matrixes have been added to the Appendix of this document.

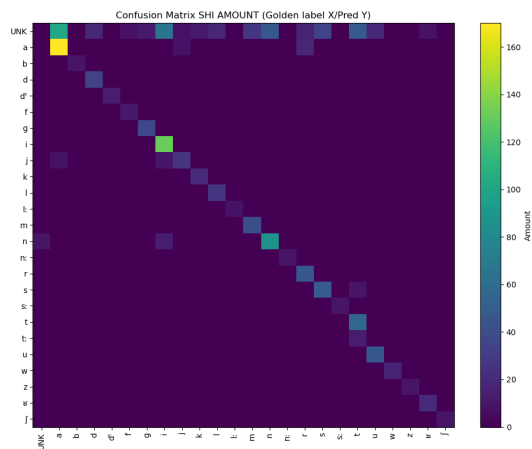
To properly read the tables: we can understand normalized by-column matrixes (normalized by prediction) as showing the characters that, when used by MFA, have not been differentiated enough in training; meanwhile tables normalized by-row (normalized by golden labeling) show us which sounds are being failed to being recognized. By-column roughly equates precision (how trust-worthy MFA is when predicting X?) and by-row roughly equates recall (can MFA recognize X?).

Looking at Figure 24 (a) and (c) one can see how many labeling pairs range between 10 and 160, making a value in the range (5,10) as minimum threshold reasonable. The reason why (c) and (d) have less symbols is that the audios from **tzm** are shorter (76 seconds in comparison to the 93 from **shi**). For context, the total amount of time trained on **kab** after handling orthography exceptions was 88006 seconds and **zgh** was 1967 seconds. Comparing the absolute occurrences at (a) and (c):

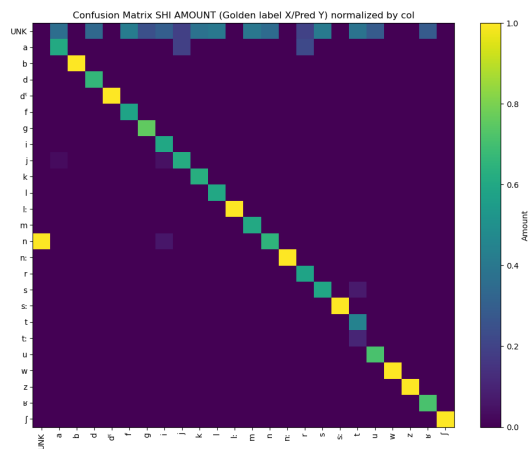
- /a/, /i/ and /n/ are the most common phones, with the only geminate appearing in **tzm** often enough being /d:/.

Looking at (b) and (d): the commonly mistaken as silence tokens in **tzm(d)** are mistaken in a similar ratio for **shi(b)**: /a/, /i/, /m/, /n/, /s/, /t/, /u/. I understand this as the model failing in similar manners for both, so let me comment what is interesting in (b):

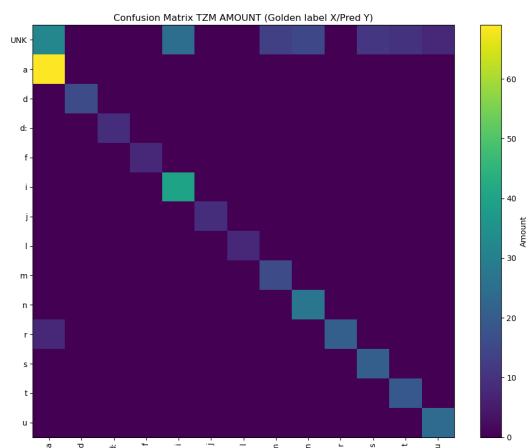
- An interesting assortment of phones are not misspredicted consistently as **UNK**: /b/, /d^h/, /l:/, /n:/, /s:/, /z/ and /ʃ/. Comparing (a) and (b): these are symbols with a very low occurrence, so it just may be there are not enough mislabeled samples to detect other patterns. There is also the possibility that; since geminates are longer and fricative/plosives are easier to differentiate when looking an audio, they are more easily learned by the MFA.
- /w/ is also well-predicted consistently and /j/ is confused as /a/. The /j/ confusion is consistent to rhe MFA performing badly with vowels/nasals, but intuitively /j/ should be confused with /i/ and /w/ should be confused with /u/.
- /t:/ is not predicted often by the model; this is consistent with having not enough samples of **zgh** that show the geminate /t:/ sound. The phone-grouping of /t:/ and /t/ makes probably both of them being predicted a bit more interchangeably.
- The overall most common mistake is for the MFA to not label (and show a high **UNK**) but when the MFA mislabels as silence it is often an /n/.



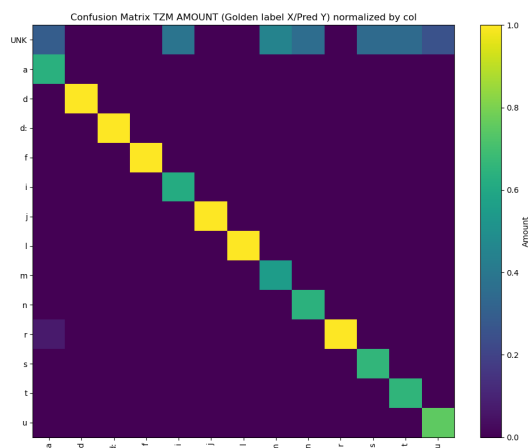
(a) SHI non-normalized.



(b) SHI normalized by column.



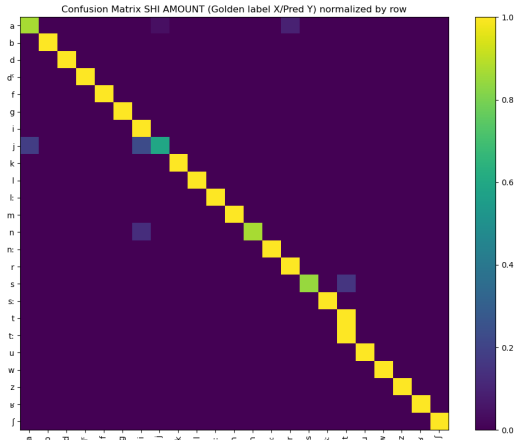
(c) TZM non-normalized.



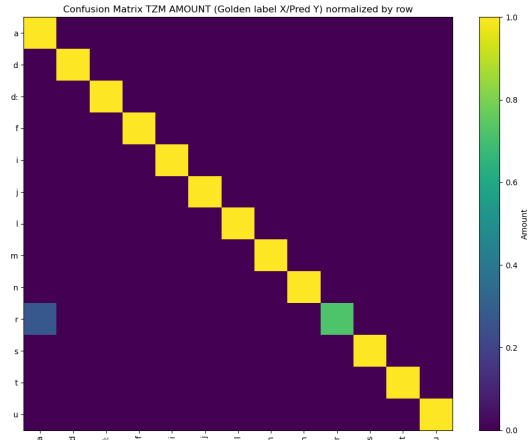
(d) TZM normalized by column.

Figure 24: Per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence AMOUNT (with a minimum of 8 occurrences).

Removing **UNK** from the table and normalizing by row (Figure 25) we can see how the MFA predicts often mistakenly a golden-labeled /r/ as /a/ in **tzm**. Rhotics' waveform is usually similar to vowels, so this error is within expectations. We see however that the reverse happens in **shi** (golden-labeled /a/ is predicted as /j/ or /r/). We see instead /n/ being sometimes predicted as /i/; which also has a similar waveform to vowels. In **shi**, we also see general issues labeling /j/, which are probably also true in **tzm** not appearing due to not enough samples. I have also graphics for the mean, median and variance (that can be seen in the Appendix) for both dialects that confirm this being a consistent and relevant; however I am omitting them here for the sake of brevity.



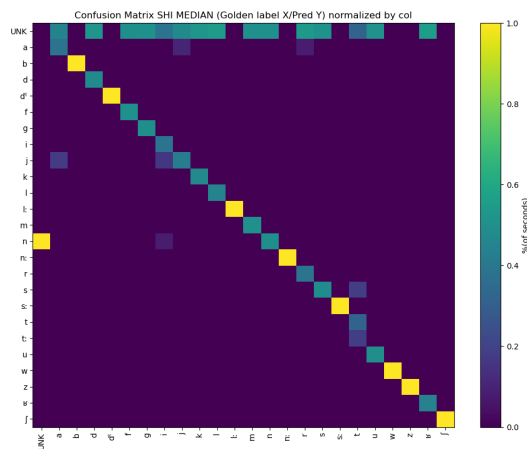
(a) SHI normalized by row.



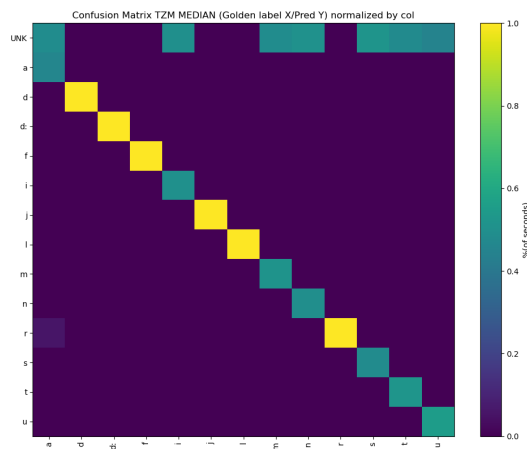
(b) TZN normalized by row.

Figure 25: Per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence AMOUNT (with a minimum of 8 occurrences) removing UNK

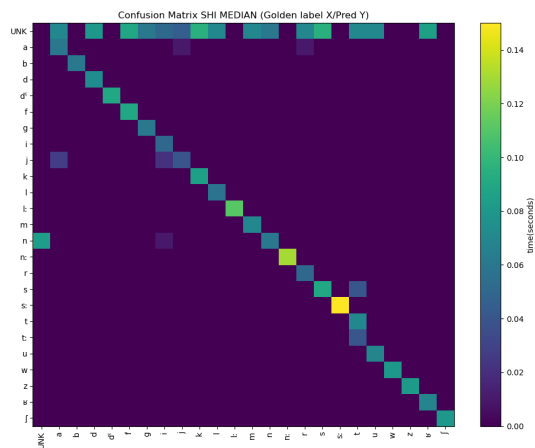
The median and variance can be visualized in Figure 26. As we can see, most of the MFA mislabels that happen often tend to also have the same length as the **UNK**; meaning the patterns for occurrences are realistic and not just fringes of a phone. Looking back at Figure 24 one gets an idea of how often these happen, in general it can be roughly understood that the model can predict more often correctly than not the boundaries. Figure 26 we can also see that the the median value for the only geminates consistently are labeled as long are /d:/ and /s:/, suggesting that /n:/ and /t:/ (mispredicted as /t/ always) are not actually longer pronunciations, contrary to the sources [9].



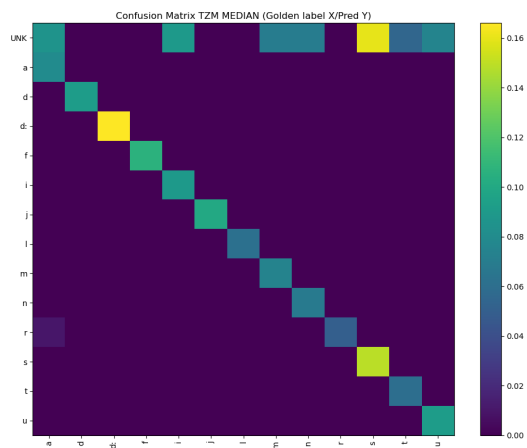
(a) SHI MEDIAN normalized by column.



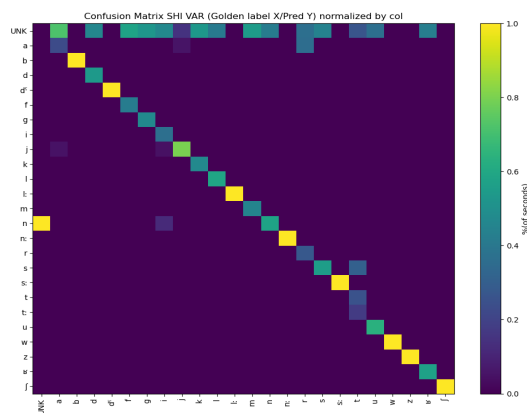
(b) TZM MEDIAN normalized by column.



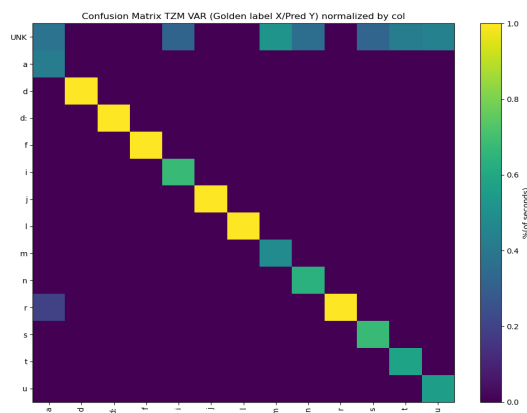
(c) SHI MEDIAN without normalizing.



(d) TZM MEDIAN without normalizing.



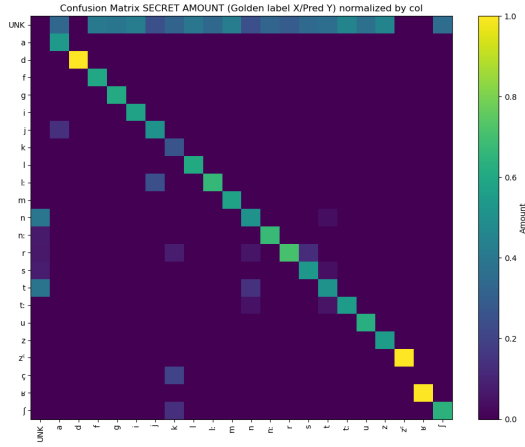
(e) SHI VAR normalized by column.



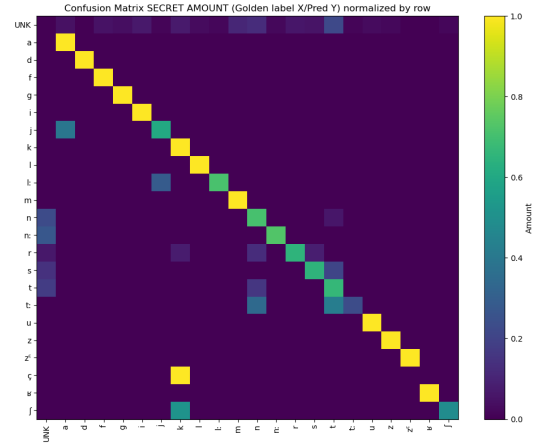
(f) TZM VAR normalized by column.

Figure 26: Per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing statistical variables (with a minimum of 8 occurrences).

9.2 Private dataset



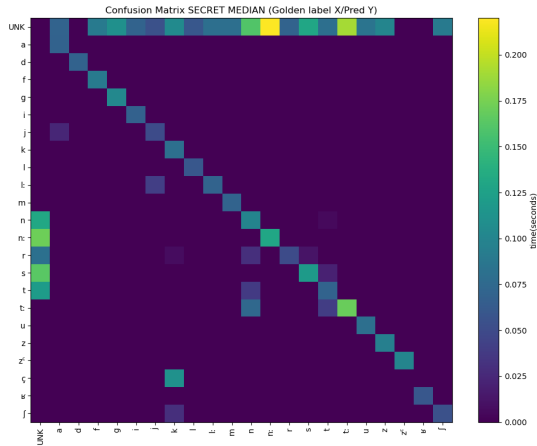
(a) Amount normalized by column.



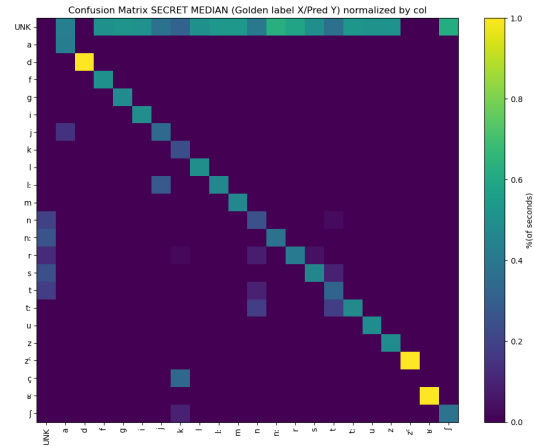
(b) Amount normalized by row.

Figure 27: Per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing total amount (with a minimum of 8 occurrences).

Figure 27 matches my manual annotation observations well; often /k/ is predicted when a /ç/ is the true label (the MFA has no way of predicting /ç/) and the transition /j/ ->/l/ having issues. It is not surprising that the golden /n/ and /t/ are often predicted as out of bounds, as the data over-represents these two in final positions. These reinforce the idea that the MFA struggles with /t/, /s/, /j/ and /n/, but we don't see the same /r/ issues as in Common Voice, probably because /r/ is less represented here. Looking at the medians in Figure 28 are once again around 0.1 seconds with clear issues regarding both identifying and predicting /n:/.



(a) Median non-normalized.



(b) Median normalized by column.

Figure 28: Per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing total amount (with a minimum of 8 occurrences) on the Private Dataset.

10 Statistical analysis per-sample

10.1 Common Voice

Looking at the figures 29,30, 31, 32,33 and 34 we have a general idea of how "roughly correct" the prediction is given a single sentence. My intention here was to compare a bit some of the intuitions I had while going through the manual analysis: I had seen very often the 1-2 word prepositions being completely mislabeled, and I had seen often as well the alignments being overall correct.

Looking at the per-sample results, all the averages fall within the range of 50 to 65% of correct seconds. When looking at the phone-level the performance is a 4-9% lower. This is within expectations given several issues with my orthography that I realized while doing manual annotation: some common realizations cannot be predicted by the MFA with the current orthography (devoicing, spirantizations, cross-word assimilations).

Sample 5 on **tzm** excluded due to having an entire word (unlabeled) at the beginning, and Sample 0 has been excluded due to the MFA refusing to generate it when not trained with the **kab** corpus (it contains "XX", and I presume /χ:/ is rare enough to not appear as a symbol in that case).

If one checks how the model that was not trained on **kab** data performs (plots added to the Appendix) one can see how MFA performs abysmally in the respective **tzm** word and phone analysis (but not so bad when excluding 1-2 length words). On **shi** the results' means are comparable (within 5% difference), but the variances are across all graphics significantly higher (PHON: 43, WORDS:150, NOSTOP: 50) in comparison to Figures 29,30, 31 (PHON:11, WORDS:50, NOSTOP: 23).

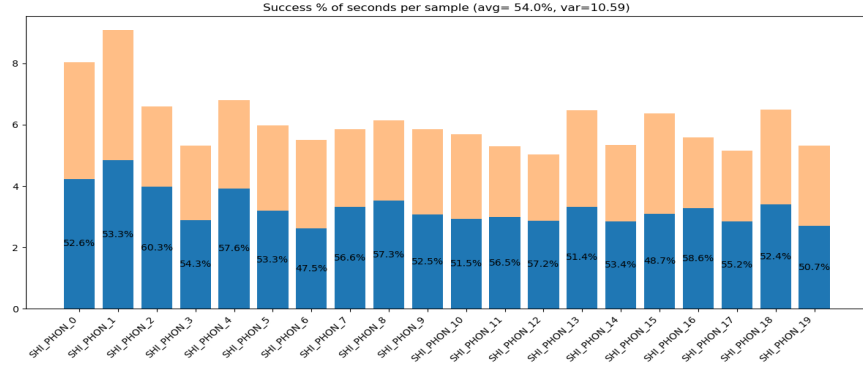


Figure 29: **shi** correctly overlap % of seconds between predicted/golden phones, per-sample

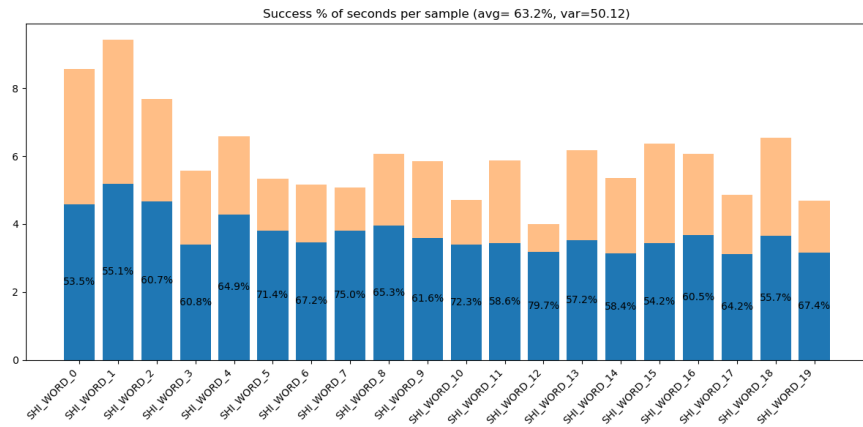


Figure 30: **shi** correctly overlap % of seconds between predicted/golden words, per-sample

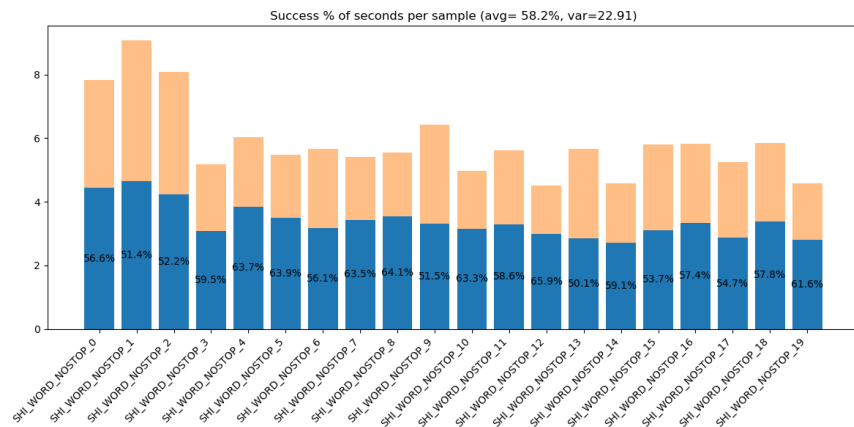


Figure 31: **shi** correctly overlap % of seconds between predicted/golden words (excluding 1-2 length words), per-sample

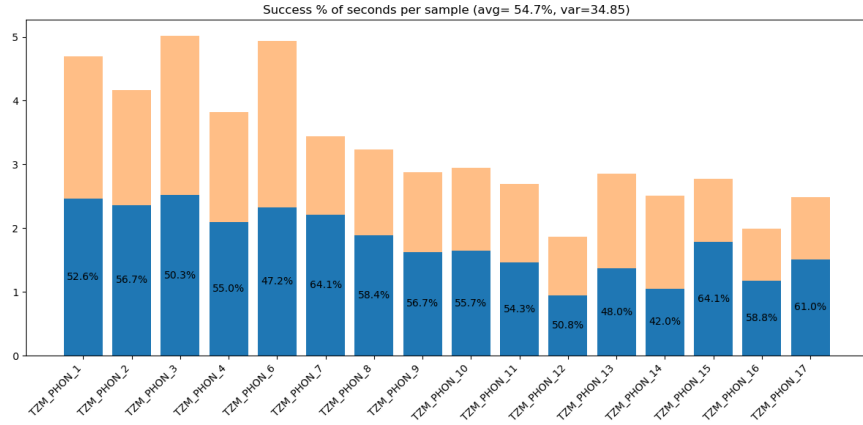


Figure 32: **tzm** correctly overlap % of seconds between predicted/golden phones, per-sample

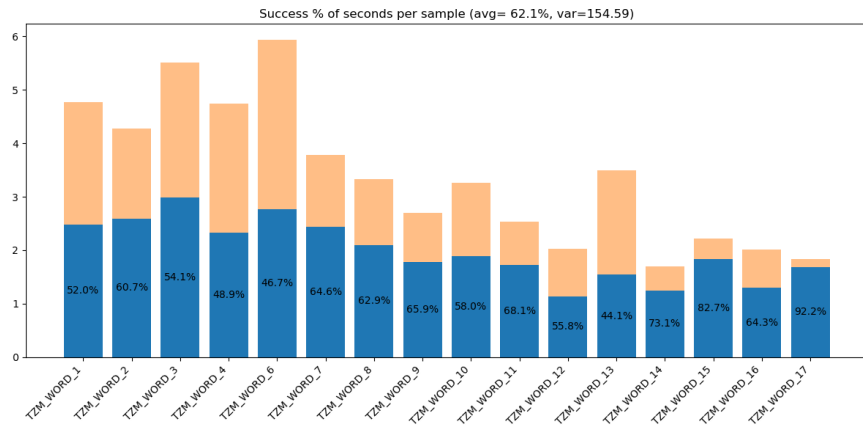


Figure 33: **tzm** correctly overlap % of seconds between predicted/golden words, per-sample

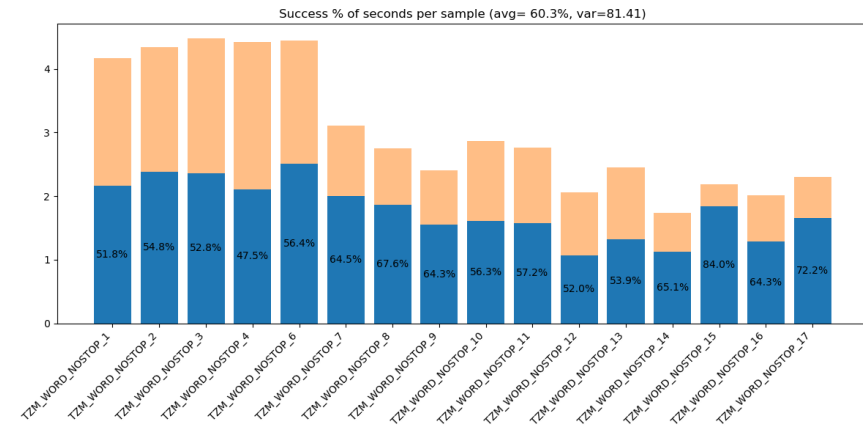


Figure 34: **tzm** correctly overlap % of seconds between predicted/golden words (excluding 1-2 length words), per-sample

10.2 Private Dataset

It is clear that the labeling is a lot more off in the private dataset in comparison to Common Voice's. Longer audios allow for more chances of a mislabel making it impossible to label properly future words. The phone and word analysis here look more similar not because the common voice samples being too good on per-word labeling; rather because here the per-word labelling is much more likely to go awry.

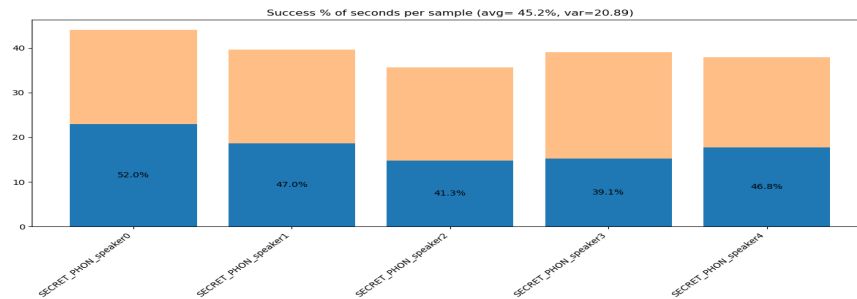


Figure 35: **shi** correctly overlap % of seconds between predicted/golden phones, per-sample

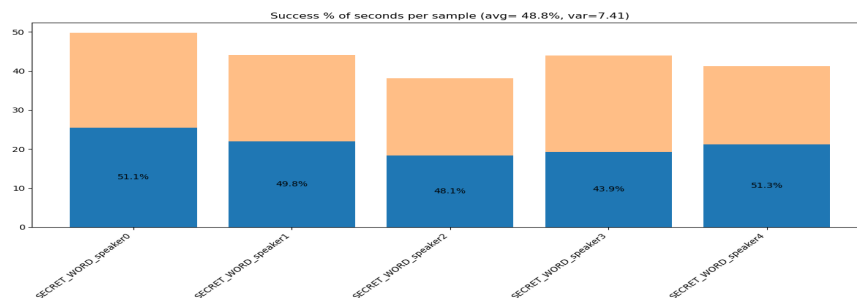


Figure 36: **shi** correctly overlap % of seconds between predicted/golden words, per-sample

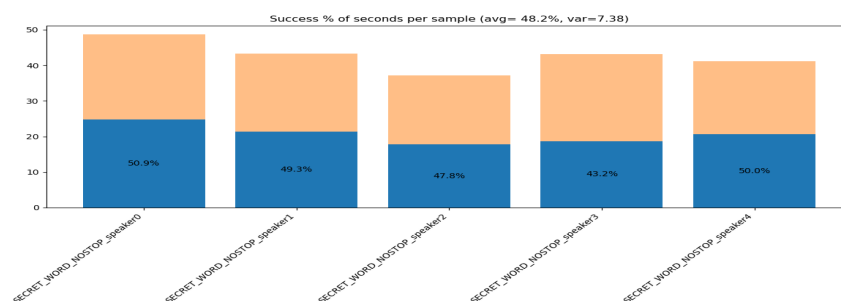


Figure 37: **shi** correctly overlap % of seconds between predicted/golden words (excluding 1-2 length words), per-sample

11 Limitations

I am not a native speaker of any Amazigh variant. I am also a computer science before a phonologist, and this may show in my approach. Reproduceability and standarization have been my focus since the beginning of this Thesis, and while Tifinagh and Berber are somewhat standardized in writing; it was very challenging to find a well-reputed source that suggests a general good approach to orthographic transcription across berber languages. This work's focus was also to try and be able to provide such a standard, but it is entirely possible I have missed spirantizations that phonologists more experienced in Berber will detect. I have mostly used [24] as my base for Linguistics; and [9] as a starting point for Tachelhit, combining it with the Wikipedia pages (and their sources) and some native speakers help.

It would be interesting to cover all the languages included under the **zgh** standard, given how some studies [10] show us that training on a series of low-resource languages is advisable and that **tzm** is described as a a continuum between **shi** and **rif** ³⁷. However finding a readily available dataset for **rif** has been too difficult, and after noticing several issues with some of the datasets I was counting on I decided to prioritize reliable data. The letter "r" is also very often vocalized ³⁸ which distances this variant of Berber further from the particularities of **shi/tzm**. It makes me suspect also that a lot of french influence may exist in **rif**. Some research exists [18], with decent reproduceability (code released without audios); however I could not find any audio dataset readily available for **rif** as of 2026.04, having to discard it for study.

The special writing of the gemminate emphatic consonants has not been corrected in-code, as it was discovered quite late in my research. It is worth noting that while most research clearly mentions how geminates and emphatic consonants, there is usually no reference of both happening simultaneously or how they should be represented.

12 Conclusions

As was suggested by the research [10], it can be seen that using languages from the same family greatly improves the performance and robustness of an MFA, specially when unseen languages come into play (as is mostly the case with **tzm**). This is a valid way to somewhat bypass the corpus limitations of a low-resource language.

I have some further considerations regarding the MFA: I initially avoided adding dialect-specific spirantizations due to seeing poor cross-comparizon between the research that mentioned them. In theory, multiple realizations of a word are fine and the MFA underlying pipeline should know which pronunciation to use. In practice the suggested multiple realizations of the preposition "n" resulted in the MFA completely mislabeling the phone that should be heard more often than not. If dialect-specific spirantizations were to be added, I would strongly advise to guarantee that a significant amount of corpus in said variant exists. The same holds true for assimilations.

13 Future work

I really focused on this thesis to try and have a correct representation of the orthography of **shi**, which meant a lot of my time was dedicated to reworking the orthographic transliteration for **zgh**. This has come at the tradeoff of less time dedicated to MFA customization itself and the evaluation

³⁷https://en.wikipedia.org/wiki/Central_Atlas_Tamazight

³⁸<https://en.wikipedia.org/wiki/Tarifit>

algorithms for the alignments.

Some suggestions regarding better evaluation:

- In general, my formulas use overlapping between golden and predicted alignments; which is not as robust a metric as Levenshtein's edit distance. I wish I had the time to properly test a function that calculates edit distance on intervals, and proper weighting for edit, insert, deletion and thresholds at which we don't consider an error. Memoization-like algorithms are always hard to program, so I decided to prioritize verifiable-yet-not-so-robust metrics. In fact one of the criticized papers [17] uses Character Error Rate which is a much more advisable metric, and the implementation is properly published, so integrating their implementation would be a sensible idea.
- In the most comparable research to mine [10] a visualization is done on the vowel space; I decided to focus my analysis more on the confusion matrixes between golden and automatic alignments, but this would be definitely a parallel analysis that may give us interesting insights. The models are also published within my repository, making these accessible for a 3rd party to compare.
- I did not provide any analysis of how much support does the MFA for specific orthography get from the corpus itself. It would be sensible to add to the analysis an estimate of how many samples of each phone are expected to be used in training. E.g. Given a binomial distribution, with a total of 150k samples where we sample 40k random samples, the /l:/ phoneme appearing in 20k rows

$$\mathbb{E}[/l : /observations] = \frac{20,000 \times 40,000}{150,000} = 5, \bar{3}$$

One could get an idea of which letters have no representation in the biggest training corpus (outside /t:/)

As is evident, getting reliable data for this project and verifying it has been the biggest challenge. Here are some suggestions regarding adding more curated data that would align with this project:

- Seeing how, as suggested [10], more variants within a family are useful; it would be interesting to add some **rif** data. In particular AMZSRD [14] contains precisely **rif** and if it had been available publicly (or privately) it would have been a valuable training corpus and useful to see which phones do not transfer well from **kab**.
- One of the **shi** datasets [15] was not added due to it being detected very late; but the person behind this dataset was reachable, a native speaker and had a linguistics background; reliability has proven an issue when obtaining datasets and this dataset inspires me trust.

In early stages of this project, my intention was to try and make the pronunciation dictionaries as comprehensive as possible. With that goal in mind I collected several sources that would allow one to build the dictionary independently from the audio. Implementing the orthographical pronunciation left me no time to do so, but it would be definitely be better practice to build the dictionary from independent sources. I noted the sources down on my codebase ³⁹, but my

³⁹https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/zgh_build_dicts.py

suggestions would be to use the wikipedia dumps on **shi**. These are readily available ⁴⁰ but require following a bit of a tutorial to process the data ⁴¹. If one gets through the hurdle, 14k pages are available. I would also explicitly not recommend using Tatoeba's data ⁴² as a lot of the data available is **kab** data mislabeled as **zgh**.

As further work to do on the dictionaries, I was a bit too conservative with phonological rules to overgenerate vocabulary. MFA describes them exclusively as the option "to over-generate a lexicon"; so it should be fine to aggressively generate all possible realizations for a phone (assimilations and realizations).

While finishing my analysis and seeing how /t:/ is a clear issue for the MFA when working on **shi**; I think a simple workaround could be advisable. One could label on slow speakers (calculated by picking the top 10% of audio_length/text_length) pronunciation of /t/ in the **kab** corpus as /t:/. While this is not ideal, the only difference between the two should be length, meaning that depending on the speaker speed some /t:/ and /t/ should overlap. It would be interesting to see whether cross-word geminate assimilation improves with this. Note this would be applicable to other geminates as well.

It would be interesting as well to orthographically add vocoids between every consonant where it is possible. I documented in https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/zgh_build_dicts.py which cases would cover that based on [9] p. 139, to ease this task on a prospective future researcher.

Last but not least, while the dataset used [16] is impressive, downloading the 40k audios may pose a challenge for users, as it implies downloading 150GB of data through the internet. After transforming it to the ASR standard (16-bit precision, .wav resampled at 16kHz) this data takes only 3GB. Having to download less than 1% of the volume of data would have certainly sped up my development of this project; and it is a hard barrier to researchers that do not have a high-speed connection. Publishing an already-resampled dataset would probably open-up research possibilities for a lot of people I would be interested in doing so (and since this data is Public Domain, no legal concerns exist over this).

14 Contributions to the topic

- **A Forced Aligner for kab** Available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/output_kab/kab_model.zip; for which currently none are available officially ⁴³
- **A Forced Aligner for zgh** Available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/output_kab/zgh_model.zip; for which currently none are available officially ⁴⁴
- **An orthographical transcription for tzm/shi** Available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/blob/main/zgh_build_dicts.py; where sev-

⁴⁰https://dumps.wikimedia.org/other/mediawiki_content_current/shiwiki/2026-02-01/xml/bzip2/

⁴¹<https://medium.com/@evan.frank/accessing-and-cleaning-bulk-wikipedia-text-data-bfde3b550474>

⁴²<https://tatoeba.org/en/downloads>

⁴³<https://mfa-models.readthedocs.io/en/latest/acoustic/index.html>

⁴⁴<https://mfa-models.readthedocs.io/en/latest/acoustic/index.html>

eral sources have been cross-referenced and cited.

- **A small manually-labeled test suite for tzm/shi** Available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/output_verified.
- **A raw-text corpus on kab/zgh.** Available at https://github.com/xarxaxdev/montreal_forced_aligner_tachelhit/tree/main/raw_sentences.

15 Statement of Originality

I hereby declare that this thesis is the product of my own work. All the assistance received in preparing this thesis and the sources used have been acknowledged. No other sources were used other than the properly cited in this document: whether through bibliography or footnotes.

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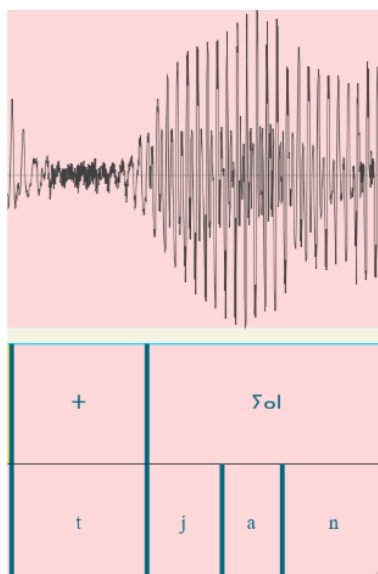
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A Appendix Manual analysis: pronunciation irregularities

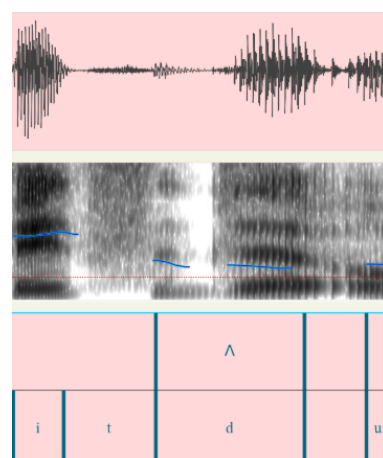
A.1 Irregularities manual analysis



(a) Automated labeling MFA. /igl:in/ is realized as /il:in/. *shi/common_voice_22_o_3*



(b) Automated labeling MFA. /t.jan/ is realized as /z.jan/. From *shi/common_voice_22_o_3*, similar behavior seeable in *shi/common_voice_22_o_6*, *shi/common_voice_22_o_11*, *shi/common_voice_22_o_12*



(c) Automated MFA Labeling. /td/ is realized /θd/. Can be lisp or **kab/rif** influence. *tzm/common_voice_22_o_4*

Figure 38: General irregularities found while manual labeling

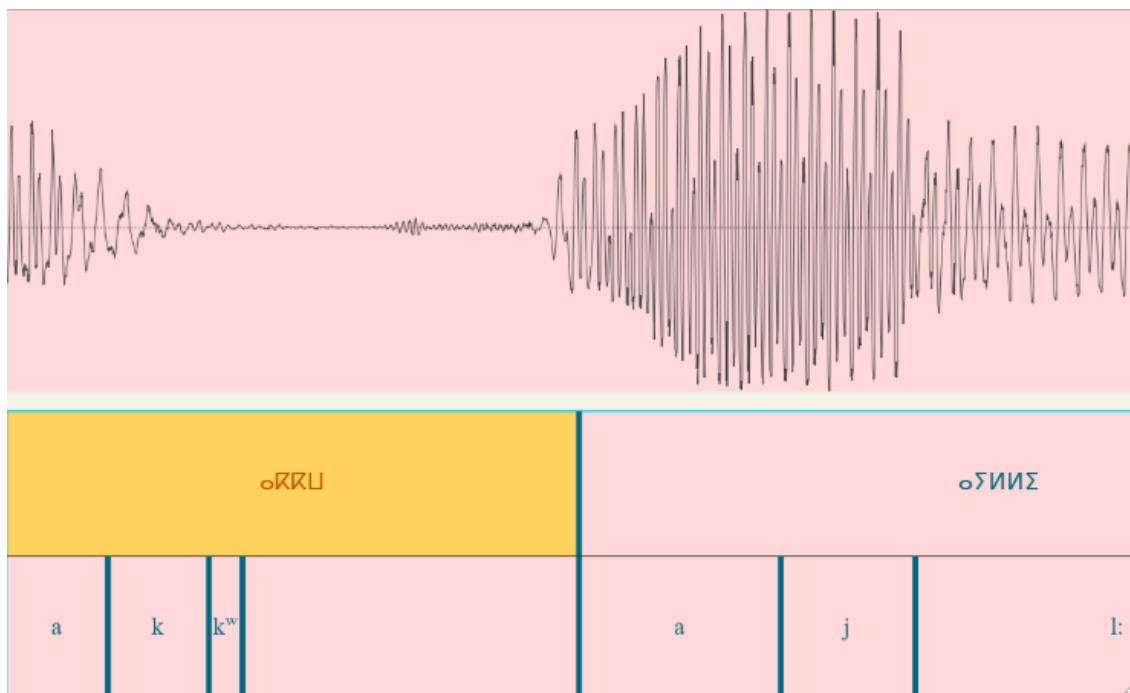


Figure 39: Automated MFA Labeling. Failed geminate labialization example, that should be corrected in the orthography. Labialized geminates writing is "oRʷ" (not "oRʷʷ"). *shi/common_voice_22_o_3*

A.2 Complete per-phone confusion matrixes of statistical measures

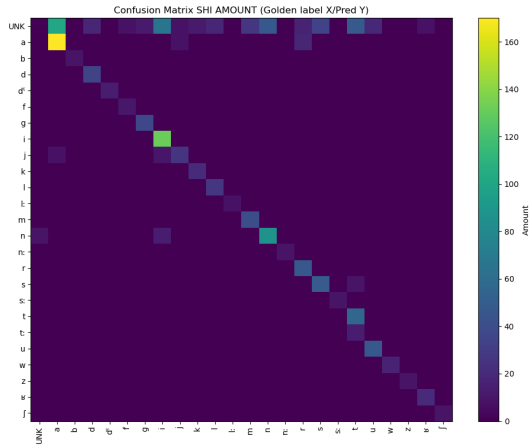


Figure 40: (a) Amount non-normalized.

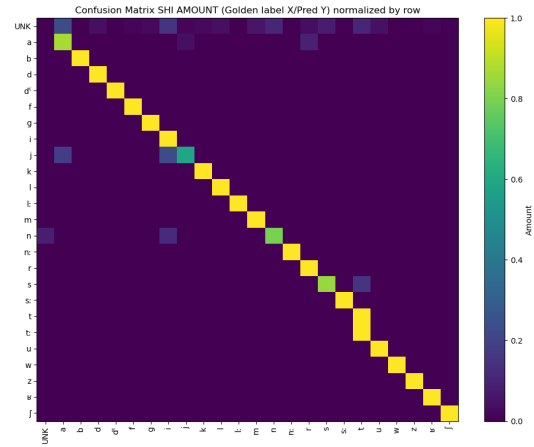


Figure 41: (b) Amount normalized by row.

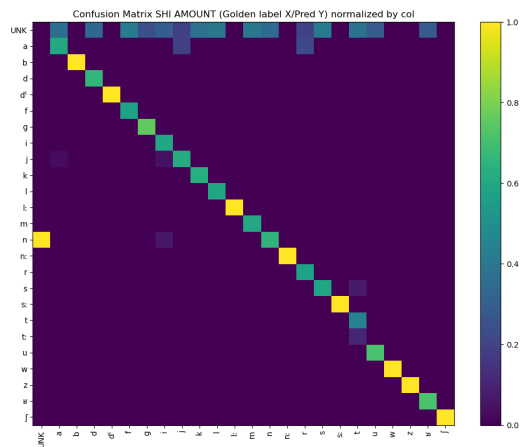


Figure 42: (c) Amount normalized by column.

Figure 43: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence AMOUNT (with a minimum of 8 occurrences).

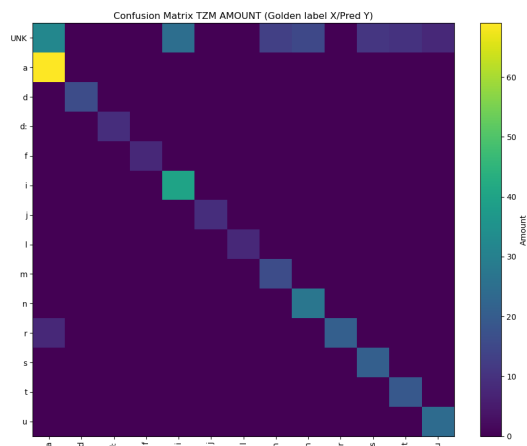


Figure 44: (a) Amount non-normalized.

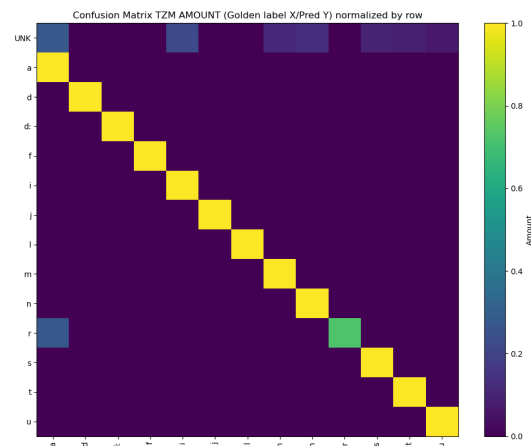


Figure 45: (b) Amount normalized by row.

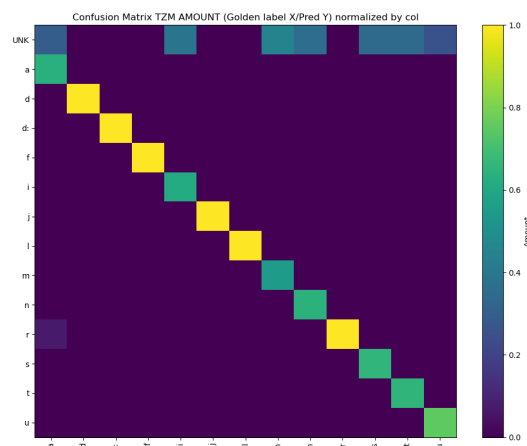


Figure 46: (c) Amount normalized by column.

Figure 47: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence AMOUNT (with a minimum of 8 occurrences).

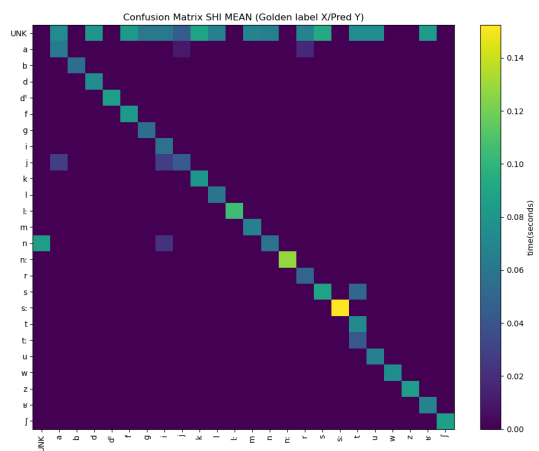


Figure 48: (a) MEAN non-normalized.

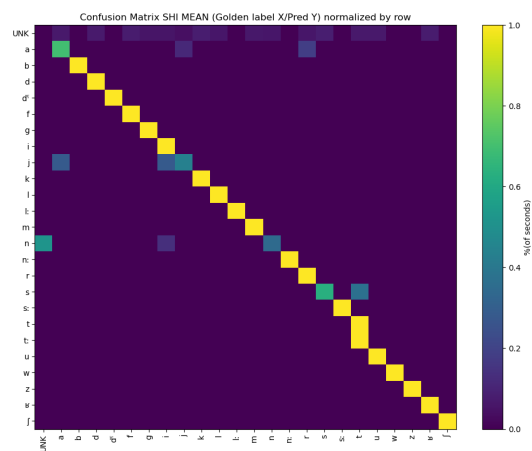


Figure 49: (b) MEAN normalized by row.

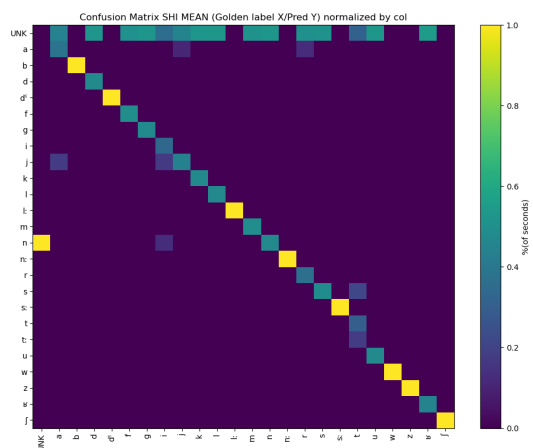


Figure 50: (c) MEAN normalized by column.

Figure 51: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEAN (with a minimum of 8 occurrences).

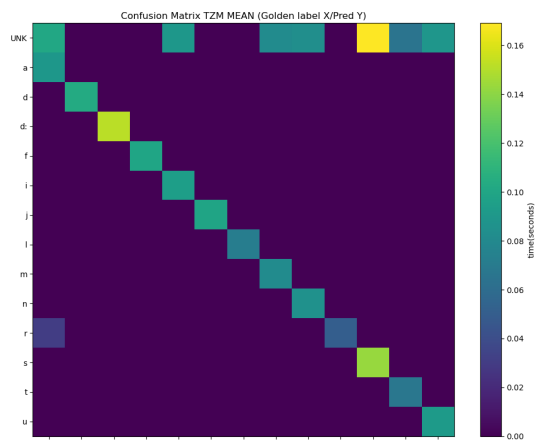


Figure 52: (a) MEAN non-normalized.

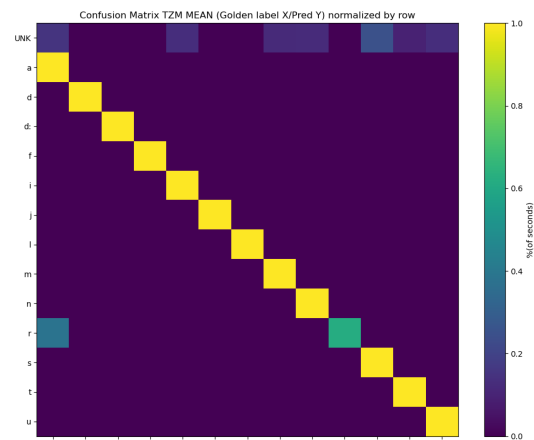


Figure 53: (b) MEAN normalized by row.

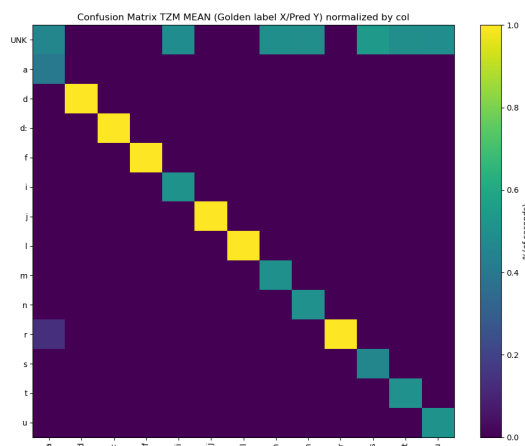


Figure 54: (c) MEAN normalized by column.

Figure 55: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEAN (with a minimum of 8 occurrences).

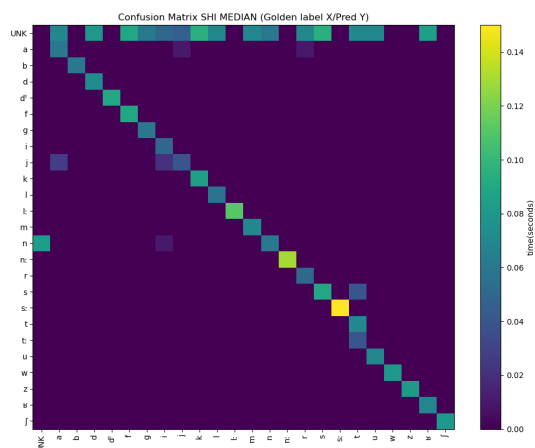


Figure 56: (a) MEDIAN non-normalized.

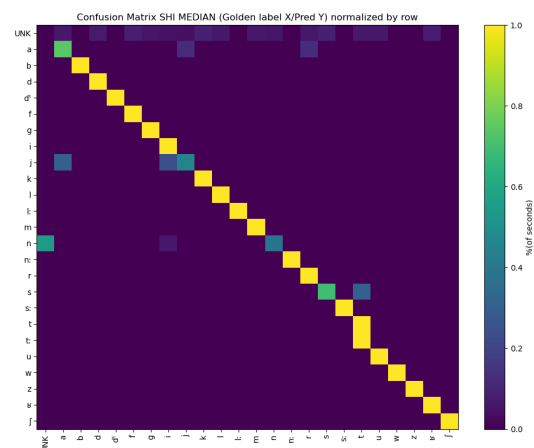


Figure 57: (b) MEDIAN normalized by row.

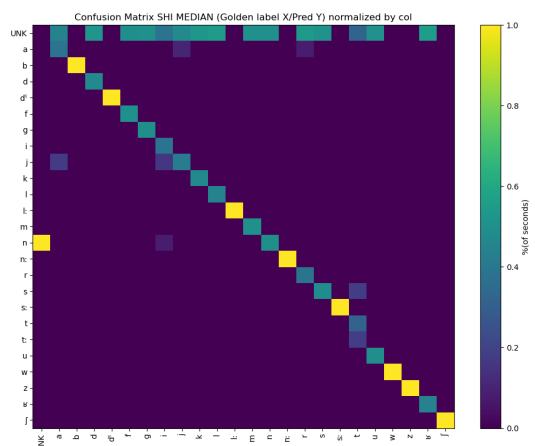


Figure 58: (c) MEDIAN normalized by column.

Figure 59: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEDIAN (with a minimum of 8 occurrences).

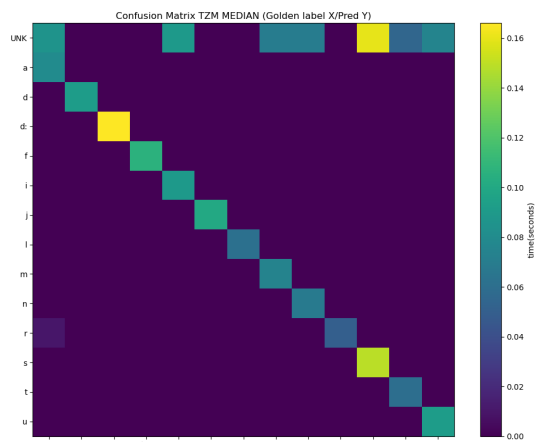


Figure 60: (a) MEDIAN non-normalized.

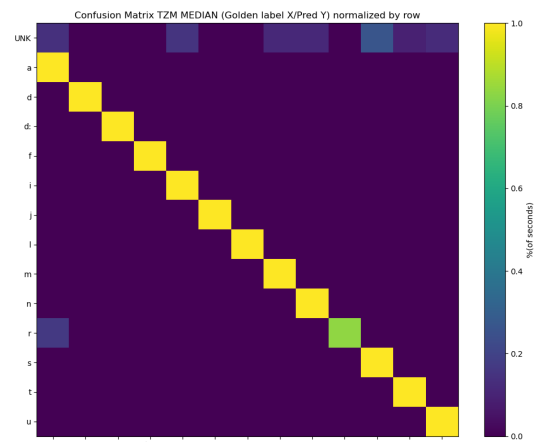


Figure 61: (b) MEDIAN normalized by row.

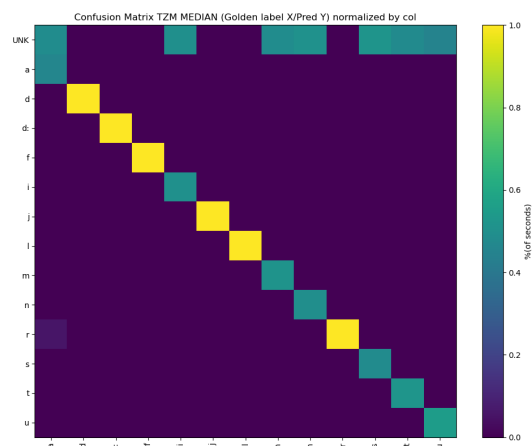


Figure 62: (c) MEDIAN normalized by column.

Figure 63: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEDIAN (with a minimum of 8 occurrences).

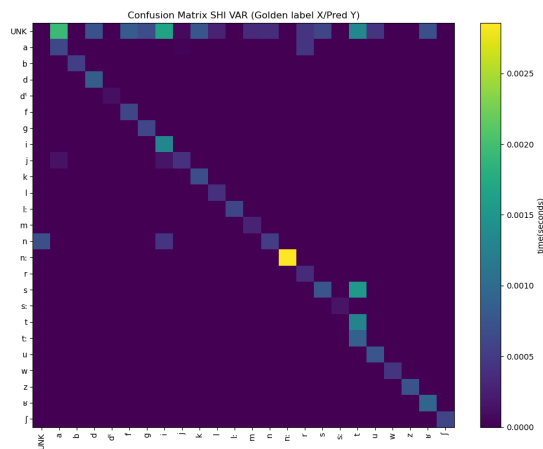


Figure 64: (a) VAR non-normalized.

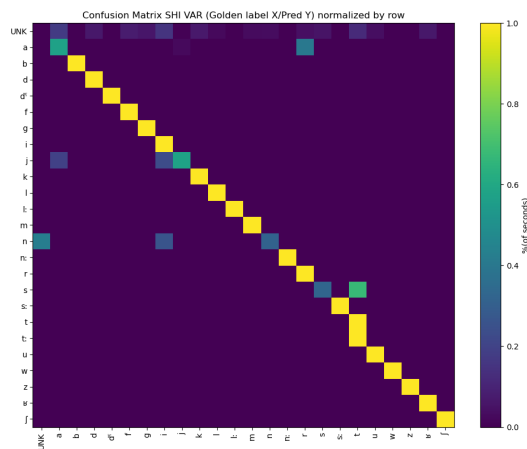


Figure 65: (b) VAR normalized by row.

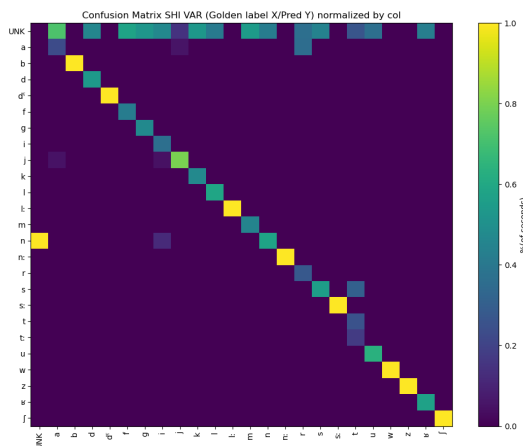


Figure 66: (c) VAR normalized by column.

Figure 67: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence VARIANCE (with a minimum of 8 occurrences).

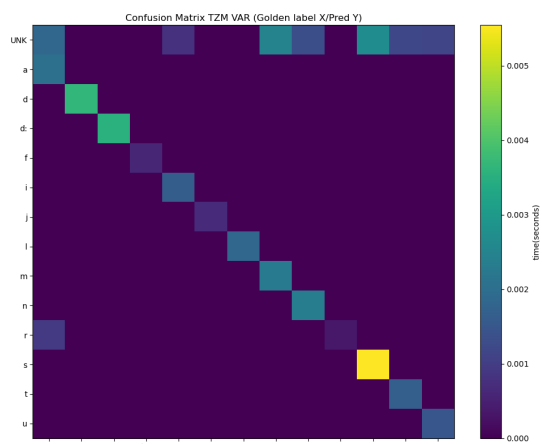


Figure 68: (a) VAR non-normalized.

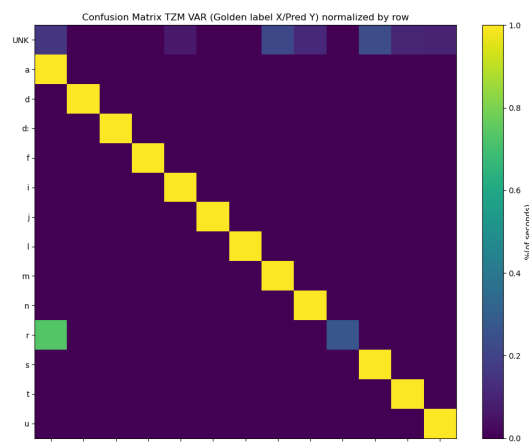


Figure 69: (b) VAR normalized by row.

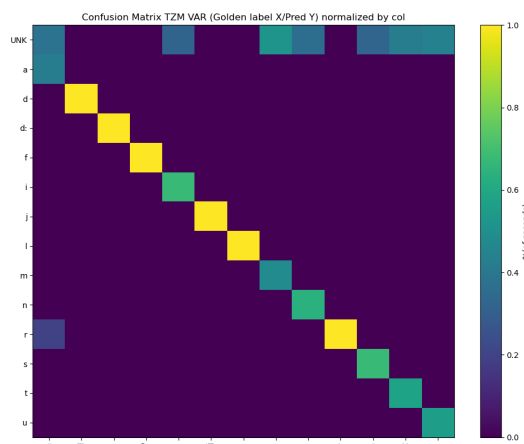


Figure 70: (c) VAR normalized by column.

Figure 71: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence VARIANCE (with a minimum of 8 occurrences).

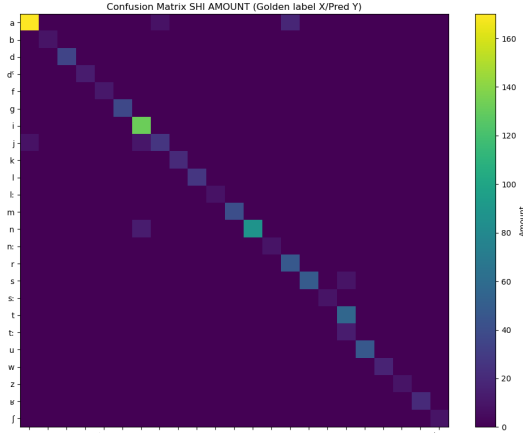


Figure 72: (a) Amount non-normalized.

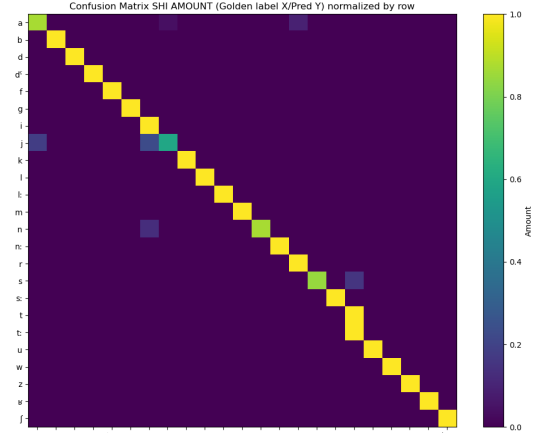


Figure 73: (b) Amount normalized by row.

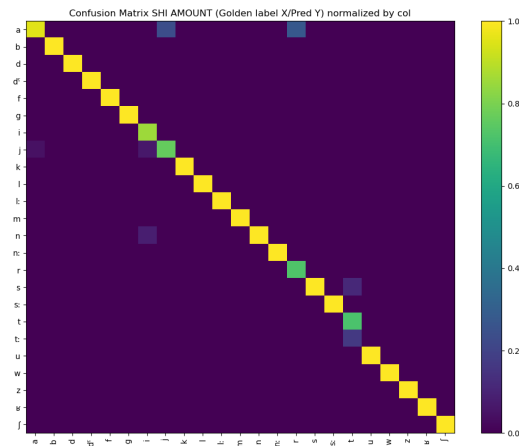


Figure 74: (c) Amount normalized by column.

Figure 75: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence AMOUNT (with a minimum of 8 occurrences) excluding UNK.

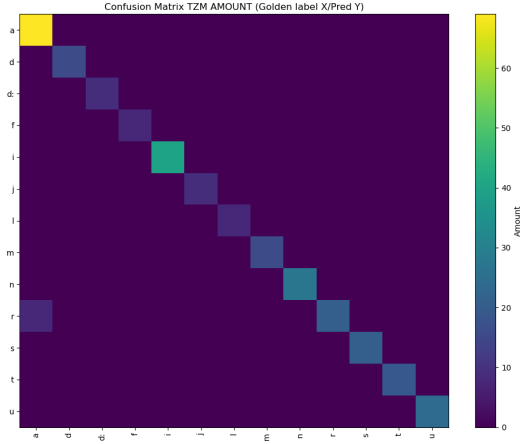


Figure 76: (a) Amount non-normalized.

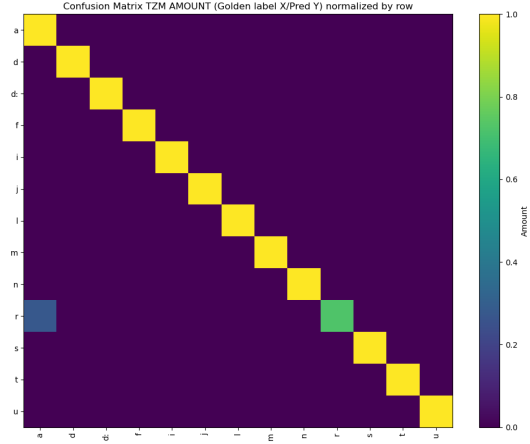


Figure 77: (b) Amount normalized by row.

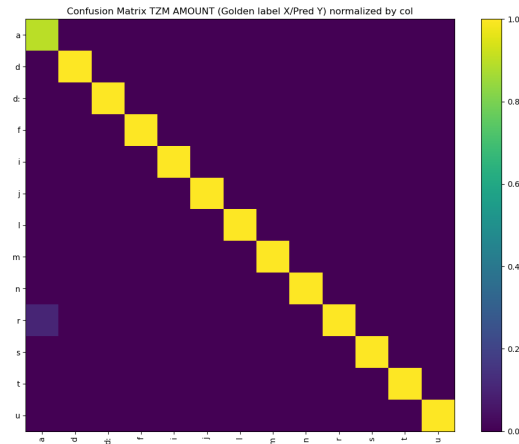


Figure 78: (c) Amount normalized by column.

Figure 79: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence AMOUNT (with a minimum of 8 occurrences) excluding UNK.

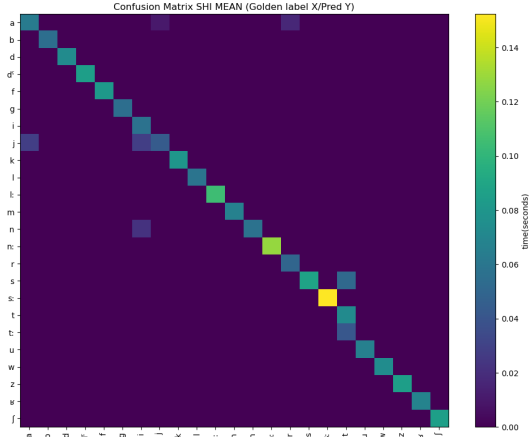


Figure 80: (a) MEAN non-normalized.

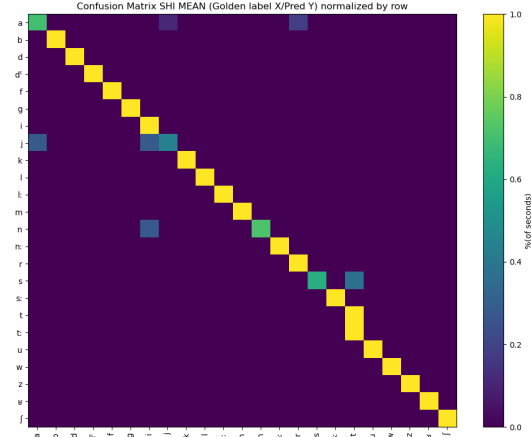


Figure 81: (b) MEAN normalized by row.

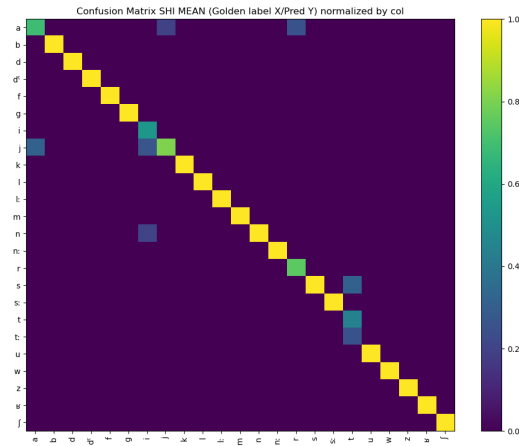


Figure 82: (c) MEAN normalized by column.

Figure 83: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEAN (with a minimum of 8 occurrences) excluding UNK.

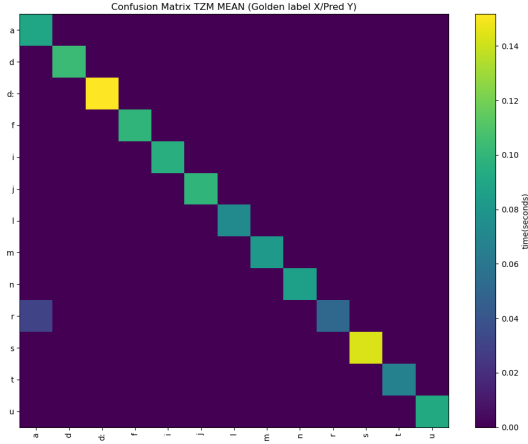


Figure 84: (a) MEAN non-normalized.

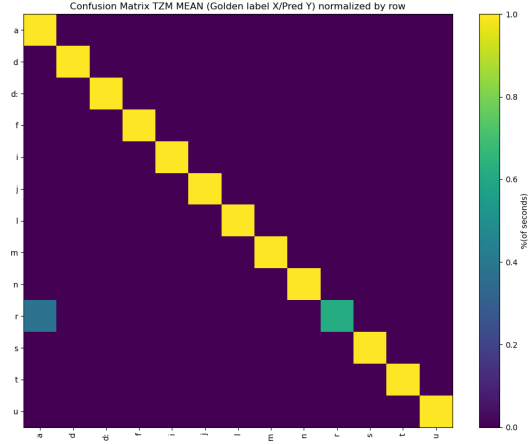


Figure 85: (b) MEAN normalized by row.

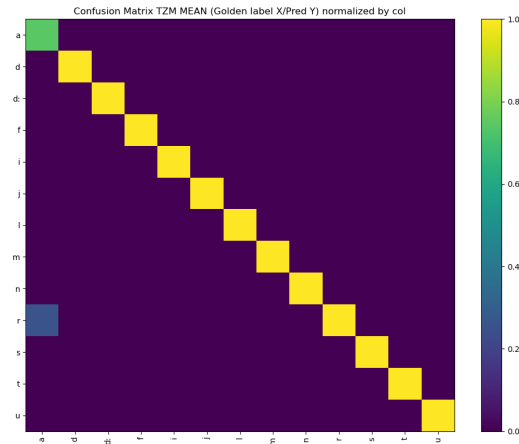


Figure 86: (c) MEAN normalized by column.

Figure 87: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEAN (with a minimum of 8 occurrences) excluding UNK.

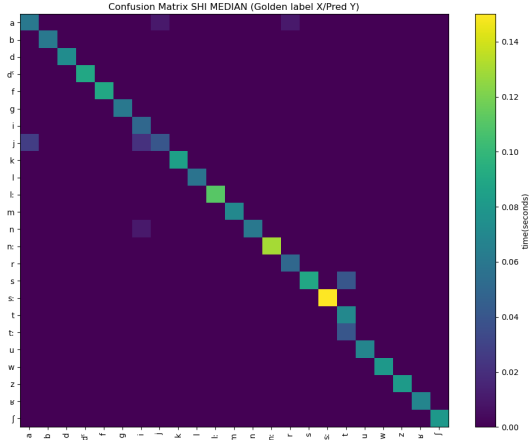


Figure 88: (a) MEDIAN non-normalized.

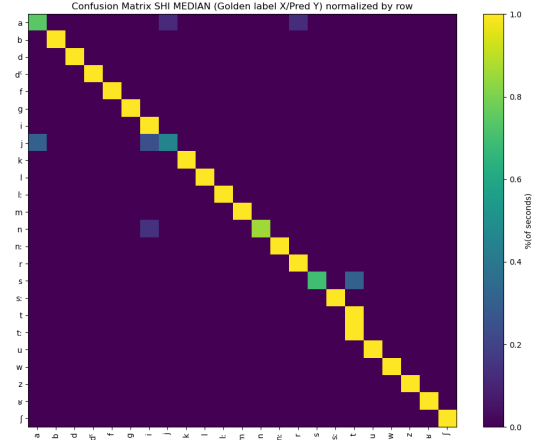


Figure 89: (b) MEDIAN normalized by row.

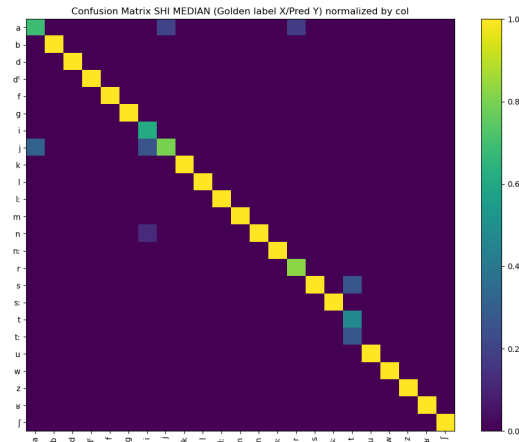


Figure 90: (c) MEDIAN normalized by column.

Figure 91: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEDIAN (with a minimum of 8 occurrences) excluding UNK.

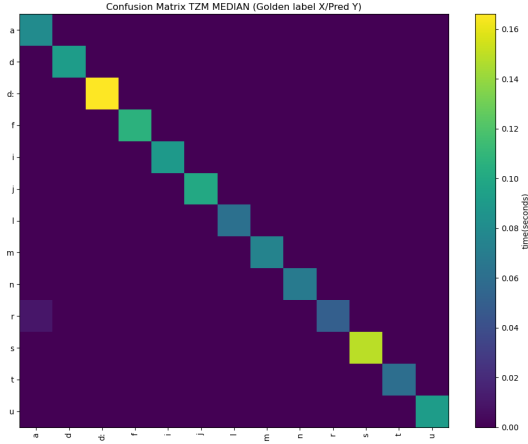


Figure 92: (a) MEDIAN non-normalized.

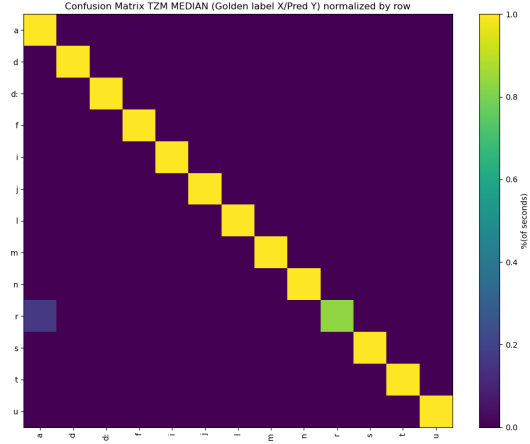


Figure 93: (b) MEDIAN normalized by row.

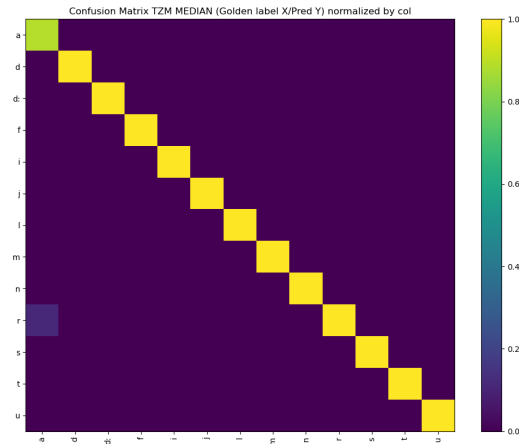


Figure 94: (c) MEDIAN normalized by column.

Figure 95: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence MEDIAN (with a minimum of 8 occurrences) excluding UNK.

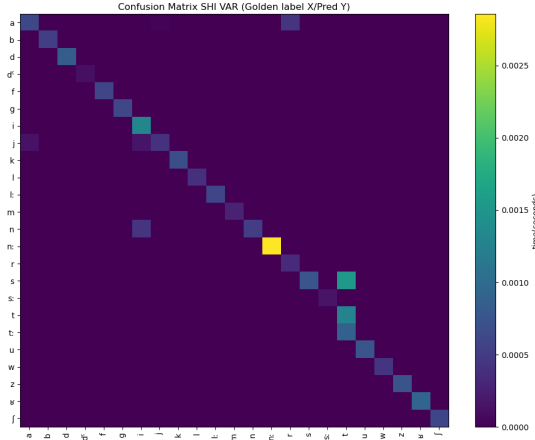


Figure 96: (a) VAR non-normalized.

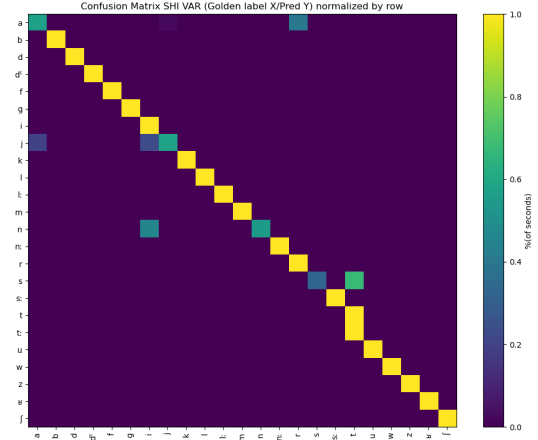


Figure 97: (b) VAR normalized by row.

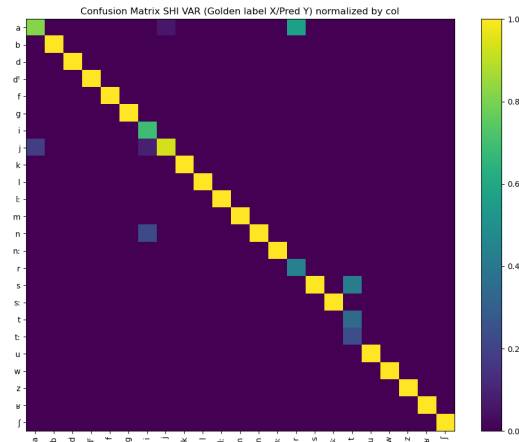


Figure 98: (c) VAR normalized by column.

Figure 99: SHI per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence VARIANCE (with a minimum of 8 occurrences) excluding UNK.

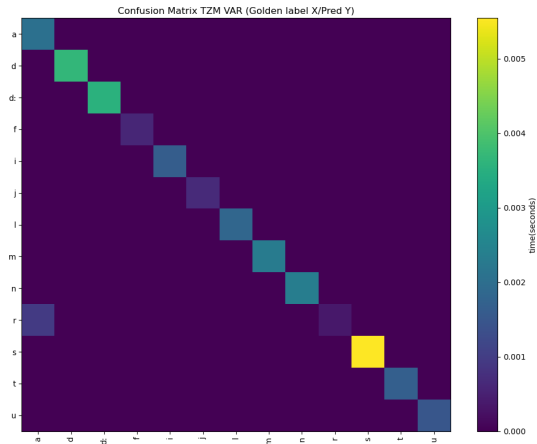


Figure 100: (a) VAR non-normalized.

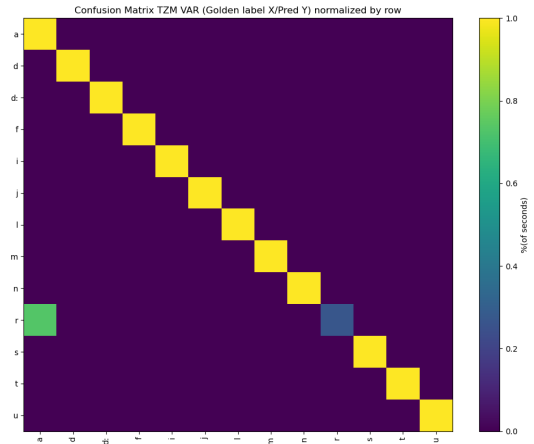


Figure 101: (b) VAR normalized by row.

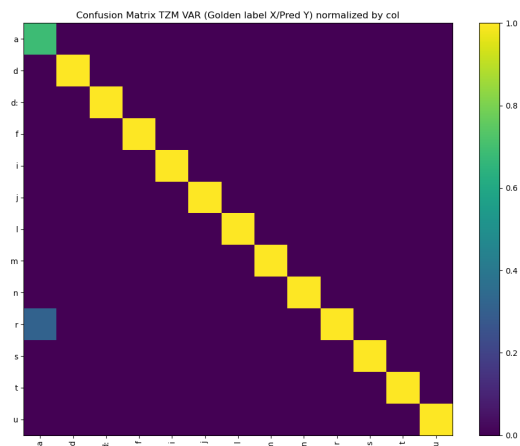


Figure 102: (c) VAR normalized by column.

Figure 103: TZM per-phone confusion matrixes of Golden(rows) vs Predicted (columns) showing the occurrence VARIANCE (with a minimum of 8 occurrences) excluding UNK.

A.3 Statistical analysis per-sample without using kab data

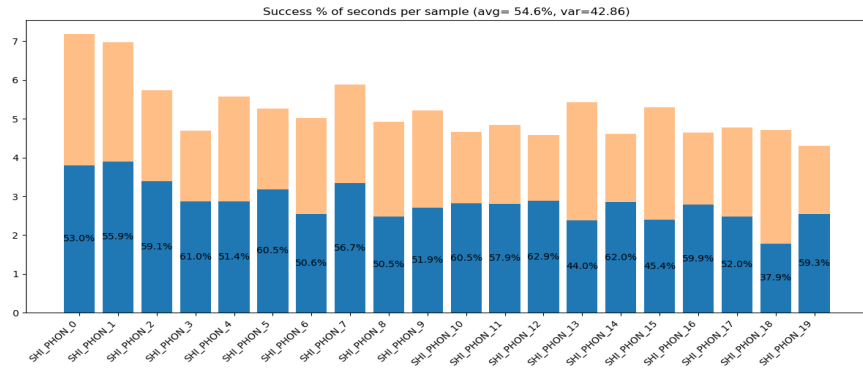


Figure 104: **shi** correctly overlap % of seconds between predicted/golden phones, per-sample with no **kab** data seen.

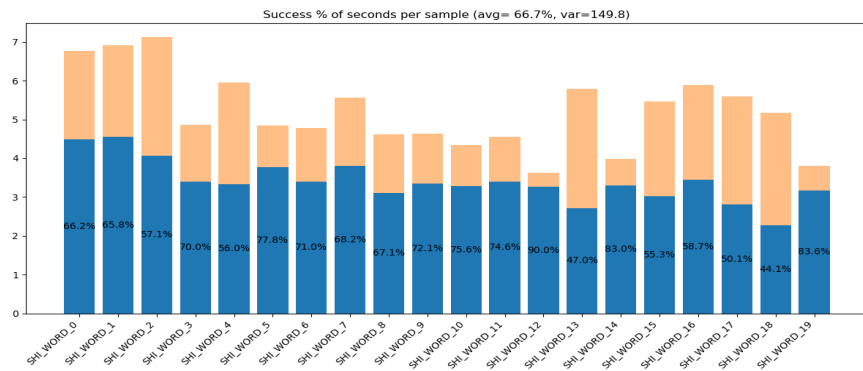


Figure 105: **shi** correctly overlap % of seconds between predicted/golden words, per-sample with no **kab** data seen.

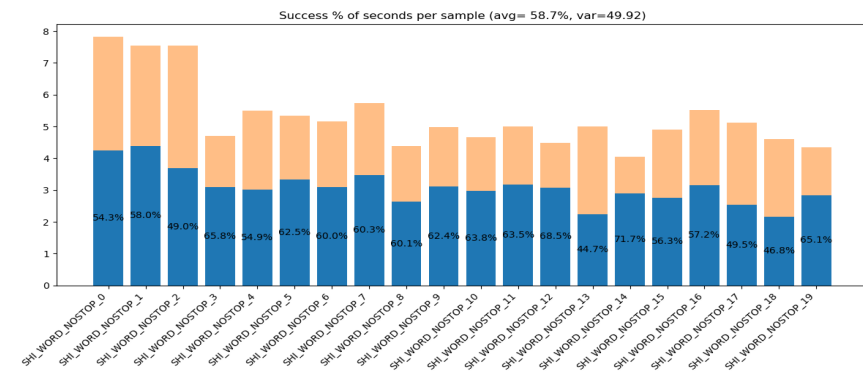


Figure 106: **shi** correctly overlap % of seconds between predicted/golden words (excluding 1-2 length words), per-sample with no **kab** data seen.

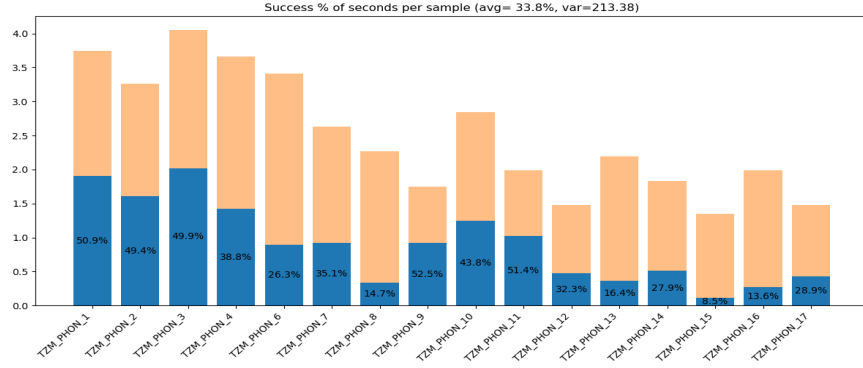


Figure 107: **tzm** correctly overlap % of seconds between predicted/golden phones, per-sample with no **kab** data seen.

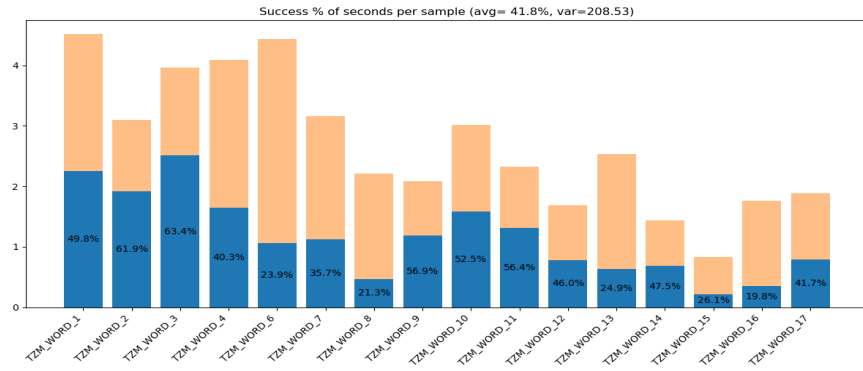


Figure 108: **tzm** correctly overlap % of seconds between predicted/golden words, per-sample with no **kab** data seen.

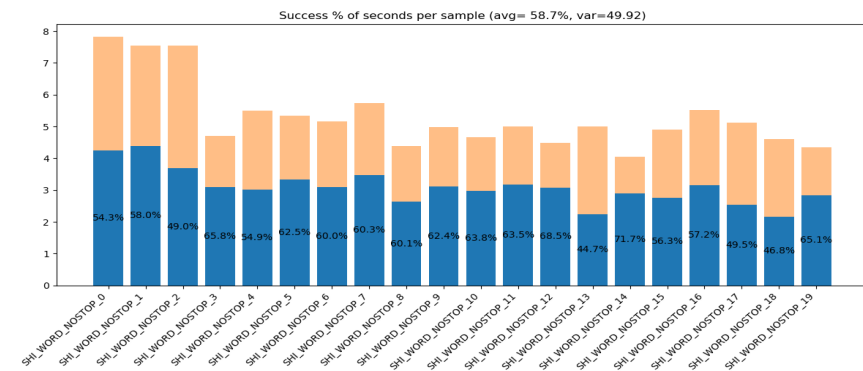


Figure 109: **tzm** correctly overlap % of seconds between predicted/golden words (excluding 1-2 length words), per-sample with no **kab** data seen.

B Abstrakt auf Deutsch

In dieser Masterarbeit werden die Herausforderungen bei der Forced Alignment mit MFA für ressourcenarme Sprachen untersucht, insbesondere die dabei auftretenden Probleme. Die Zielsprache war eine Berbersprache: Tachelhit (oder Shilha, ISO-Code: **shi**). Berbersprachen weisen unterschiedliche Grade an Online-Repräsentation auf, und für **shi** gibt es besonders wenige verlässliche Audioquellen. In dieser Arbeit werden mehrere Datensätze sowie die dabei festgestellten Probleme dokumentiert. Obwohl die Schreibung der Berbersprachen relativ standardisiert ist, ist die Aussprache der verschiedenen Dialekte weit weniger gefestigt. Im Rahmen dieser Arbeit werden die verschiedenen vorhandenen Ressourcen verglichen, um eine Orthographie zu erstellen, die eine Umwandlung von Text zu IPA in **shi/tzm** sowie separat in **kab** ermöglicht. Um die Leistung des MFA zu verbessern, wird **kab** – eine weitere Berbersprache mit deutlich mehr ASR-Ressourcen – als Basismodell verwendet, das anschließend auf **zgh** (eine standardisierte Form von **shi**) feinabgestimmt (finetuned) wird. Meine Analyse zeigt, dass die Verwendung von **kab** als Grundlage für eine bessere Forced Alignment auf **shi** empfehlenswert ist, und dass dies besonders entscheidend ist bei der Ausrichtung auf einen anderen, vom Modell nahezu ungesesehenen Dialekt, **tzm**, welcher Teil des **zgh**-Standards ist (der sowohl **tzm** als auch **shi** umfasst). Zusätzlich wird ein Vergleich zwischen Phon- und Wortebenen-Alignment durchgeführt.